A quartic cover of 37a1 and its regular S_4-closure

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Abstract

The conductor-37 elliptic curve $E\colon \xi^2+\xi=\lambda^3-\lambda$ admits a degree-4 map $E\to\mathbb{P}^1_y$ whose branch locus is governed by the irreducible cubic $c(y)=256y^3+411y^2+165y+32$. We construct the regular S_4-closure X of this cover and establish the isogeny

$$\mathrm{Jac}(X)\sim_{\mathbb{Q}}\mathrm{Jac}(X_A)\times E_{\mathrm{res}}^2\times E^3\times Q^3,$$

where $X_A\colon w^2=-y(y-1)c(y)$ is a genus-2 curve, E_{res} is the resolvent elliptic quotient of conductor 1147, and $Q=\mathrm{Prym}(Y/E_{\mathrm{res}})$ is an abelian threefold. The splitting field of $c(y)$ is the unique cubic subextension of the ray class field of $\mathbb{Q}(\sqrt{-111})$ of modulus $\mathfrak{p}_3^2$; equivalently, it is both the source of a weight-1 cusp form of level 999 and the 2-torsion field of $\mathrm{Jac}(X_A)$. We determine the bad reduction of X_A at 31 and 37, describe the 2-adic and 3-adic cluster pictures of the branch roots, and prove that the wild part of the conductor at 3 is exactly $\delta_3=1$. Thus a single branch cubic links an explicit quartic cover, a cubic ray class field, and the local arithmetic of a genus-2 Jacobian.

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1 Introduction

When an elliptic curve E over $\mathbb{Q}$ admits a morphism to $\mathbb{P}^1$ of small degree, the Galois closure of the corresponding function field extension carries a rich arithmetic structure. The quotient curves of this closure, together with the Kani–Rosen idempotent decomposition of its Jacobian [1], yield a family of abelian varieties whose conductors, endomorphism algebras, and L-functions are controlled by the branch data. In practice, however, making every piece of this picture explicit requires computing the regular closure, identifying each quotient, and matching each isotypic factor to a concrete abelian variety — a programme that quickly becomes unwieldy as the degree grows.

In this paper we carry out this programme for the degree-4 map defined by

$$\Pi(\lambda,y)=\lambda^4-\lambda^3-(2y+1)\lambda^2+\lambda+y(y+1) \tag{1}$$

and the cubic

$$c(y) = 256y^3 + 411y^2 + 165y + 32. \tag{2}$$

After the substitution $y = \lambda^2 + \xi$, the equation $\Pi(\lambda, y) = 0$ becomes the Weierstrass model $\xi^2 + \xi = \lambda^3 - \lambda$ of the conductor-37 elliptic curve (Cremona label 37a1), and the function $y = \lambda^2 + \xi$ gives a degree-4 cover $E \to \mathbb{P}^1_y$ whose finite branch locus is $\{0, 1\} \cup \{c(y) = 0\}$.

What makes this example particularly interesting is that the single cubic $c(y)$ simultaneously governs several apparently unrelated arithmetic objects: its splitting field yields a weight-1 modular form via class field theory, the hyperelliptic curve $w^2 = -y(y-1)c(y)$ contributes a genus-2 Jacobian to the regular closure, and the cubic resolvent of the quartic produces a second elliptic curve. The constructions involved — quartic resolvents, Jacobian decompositions, Prym varieties, ray class fields — are all standard (see [1–7] for general background), but this cover has the unusual feature that the quotient geometry, the class-field-theoretic cubic, the weight-1 Artin form, and the local branch-root data can all be tracked explicitly from the same branch polynomial.

The contribution of the paper is therefore explicit rather than abstract. The works of Kani–Rosen [1] and Carocca–Rodríguez [6] provide the rational-idempotent formalism for Jacobians with group actions, Paulhus [7] studies decomposition patterns for curves with extra automorphisms, and Mumford's and Lange–Ortega's work [3,8] develops the geometric theory of Prym varieties. What is new here is that one regular S_4-quartic cover can be carried through all of these frameworks in closed form, and that the resulting package is large enough to exhibit several phenomena simultaneously: a nontrivial genus-2 factor, a split Prym surface, a cubic ray class field and weight-1 form, and a wild prime whose contribution can still be isolated exactly. In that sense the example is meant to be informative rather than anecdotal.

The paper should therefore be read as a fully worked arithmetic example with a transferable method. Starting from one explicit quartic cover, we recover the regular S_4-closure and its quotients, apply the rational-idempotent method to decompose the Jacobian, identify the cubic splitting field by class-field-theoretic means, and then feed the p-adic root data into the conductor formulas for the genus-2 quotient. The point is not a new abstract decomposition theorem, but a complete template for the class of regular S_4-quartic covers whose branch cubic and resolvent geometry remain explicit.

The logical status of the results is as follows. The structural assertions of the paper — the S_4-closure, the quotient curves, the Jacobian decomposition, the class-field-theoretic description of the splitting field, and the exact wild contribution $\delta_3 = 1$ at the prime 3 — are proved in the text from standard geometric, representation-theoretic, and local-arithmetic input. The companion scripts are used only for finite-field point counts, Frobenius tables, and independent audit files for the explicit calculations recorded in Sections 5 and 6.

1.1 Summary of results

We show that $\Pi(\lambda, y)$ has Galois group S_4 over $\mathbb{Q}(y)$, and that the regular closure X has genus 16 (Theorem 3.2). The relevant quotients are the genus-2 curve $X_A = X/A_4$ given by $w^2 = -y(y-1)c(y)$, the genus-4 curve $X_V = X/V_4$, and the resolvent elliptic curve $E_{\text{res}} = X/D_4$, which turns out to be birational to the curve $w^2_{\text{res}} = x^3_{\text{res}} - 16x^2_{\text{res}} - 64x_{\text{res}} + 1040$ of conductor $1147 = 31 \cdot 37$ (Proposition 4.2). The map $X_V \to X_A$ is an étale cyclic triple cover (Theorem 3.4).

Using the S_4-representation structure on $H^0(X, \Omega^1_X)$, we establish the isogeny

$$\mathrm{Jac}(X) \sim_{\mathbb{Q}} \mathrm{Jac}(X_A) \times E^2_{\text{res}} \times E^3 \times Q^3,$$

where $Q = \mathrm{Prym}(Y/E_{\text{res}})$ is an abelian threefold with $Y = X/C_4$ (Theorem 4.7). The Prym surface $P = \mathrm{Prym}(X_V/X_A)$ satisfies $P \sim_{\mathbb{Q}} E^2_{\text{res}}$ and has endomorphism algebra $M_2(\mathbb{Q})$ (Proposition 4.10).

The genus-2 factor turns out to be the most delicate part of the decomposition. We record the Frobenius characteristic polynomials of $\mathrm{Jac}(X_A)$ at the good primes 5, 13, and 17, and use additional good primes up to 53 as irreducibility witnesses for the local quartic factors (Proposition 6.1). These data provide a concrete arithmetic profile for the genus-2 factor and contrast sharply with the explicitly split Prym surface E_{res}^2.

On the arithmetic side, we show that the splitting field L of $c(y)$ sits inside the ray class field of $K = \mathbb{Q}(\sqrt{-111})$ of modulus $\mathfrak{p}_3^2$, where $\mathfrak{p}_3$ is the unique prime above 3 (Theorem 5.4). The ray class group $\mathrm{Cl}_{\mathfrak{p}_3^2}(K)$ is cyclic of order 24, and L is its unique cubic subextension. The Artin representation of $L/\mathbb{Q}$ gives a weight-1 cuspidal newform of level $999 = 3^3 \cdot 37$, whose Hecke eigenvalues at unramified primes are $a_p = \#\{y \in \mathbb{F}_p : c(y) = 0\} - 1$ (Theorem 5.7).

We further show that the elliptic quotients E and E_{res} are not $\mathbb{Q}$-isogenous and that the product decomposition forces $\rho(\mathrm{Jac}(X)_{\overline{\mathbb{Q}}}) \geq 16$ (Proposition 6.6 and Corollary 6.7). At the two tame primes of bad reduction for X_A, the nodal singularities have opposite splitting types: the node at 31 is split while the node at 37 is non-split (Proposition 6.12). The 2-torsion field $\mathbb{Q}(\mathrm{Jac}(X_A)[2])$ coincides with L, and the mod-2 Galois image is the small group $S_3 \subset \mathrm{Sp}(4, \mathbb{F}_2) \cong S_6$ of order 6 (Theorem 6.17).

We also determine the $\mathbb{F}_2[S_3]$-module structure of the 2-torsion: the Weil pairing admits a Galois-invariant orthogonal splitting into two hyperbolic planes, one fixed and one carrying the standard representation of $S_3 \cong \mathrm{GL}(2, \mathbb{F}_2)$ (Theorem 6.19). At the semistable primes, the component groups and Tamagawa numbers of $\mathrm{Jac}(X_A)$ match those of E_{res} exactly (Proposition 6.21).

At the two wild primes, the p-adic cluster pictures of the branch roots are determined explicitly (Proposition 6.13). At $p = 3$, the three roots of $c(y)$ coalesce to a single cluster at fractional depth $3/2$, requiring a totally ramified cubic extension of $\mathbb{Q}_3$ to resolve; the hyperelliptic conductor formula then gives the exact wild contribution $\delta_3 = 1$ to the conductor at 3 (Theorem 6.15). At $p = 2$, two of the three roots pair with the rational roots 0 and 1 in balanced clusters of depth 5, while the third root is isolated at valuation -8.

1.2 Organization

In Section 2 we recover the elliptic model of the quartic and study the attached cubic field. Section 3 constructs the regular S_4-closure and its quotient curves. In Section 4 we establish the Jacobian decomposition via representation-theoretic methods and determine the endomorphism algebra of the Prym surface. Section 5 is devoted to the ray class field and the associated weight-1 modular form. In Section 6 we study the arithmetic of the genus-2 factor: Frobenius data, automorphic structure, Picard-number bounds, local conductor calculations, the $\mathbb{F}_2$-module structure of the 2-torsion, and numerical verification. Section 7 records several further directions suggested by the computation.

2 The quartic cover

In this section we recover the elliptic model of the quartic and extract the cubic field attached to the finite branch locus.

2.1 The elliptic model

Theorem 2.1. *Under the change of variables*

$$\xi = y - \lambda^2,$$

the affine curve $\Pi(\lambda, y) = 0$ *becomes*

$$E: \quad \xi^2 + \xi = \lambda^3 - \lambda. \tag{3}$$

Its smooth projective model is an elliptic curve over $\mathbb{Q}$ *with minimal discriminant* 37*, j-invariant* 110592/37*, and Cremona label* 37a1*.*

Proof. Treating Π as a quadratic in y gives

$$y^2 + (1 - 2\lambda^2)y + \lambda^4 - \lambda^3 - \lambda^2 + \lambda = 0.$$

Substituting $y = \lambda^2 + \xi$ gives

$$\xi^2 + \xi = \lambda^3 - \lambda.$$

This is a minimal Weierstrass model with the stated invariants [9, 10]. □

The rational function

$$y = \lambda^2 + \xi \in \mathbb{Q}(E) \tag{4}$$

defines a morphism

$$\varphi : E \longrightarrow \mathbb{P}^1_y.$$

Proposition 2.2. *The map* φ *has degree* 4*. It is totally ramified of index* 4 *above* $y = \infty$*, and has exactly five finite simple ramification points. The finite branch locus is*

$$\{0, 1\} \cup \{c(y) = 0\},$$

where $c(y)$ *is given by* (2)*. Equivalently,*

$$\mathrm{Disc}_\lambda \Pi(\lambda, y) = -y(y - 1)c(y).$$

Proof. At infinity, $\mathrm{ord}_O(\lambda) = -2$ and $\mathrm{ord}_O(\xi) = -3$, so $\mathrm{ord}_O(y) = -4$; hence $\deg \varphi = 4$ and $y = \infty$ is totally ramified.

Differentiating (3) gives $(2\xi + 1)\, d\xi = (3\lambda^2 - 1)\, d\lambda$, whence

$$dy = \frac{2\lambda(2\xi + 1) + 3\lambda^2 - 1}{2\xi + 1}\, d\lambda.$$

The ramification condition $dy = 0$ requires $2\lambda(2\xi+1)+3\lambda^2-1 = 0$, that is, $\xi = -(3\lambda^2+2\lambda-1)/(4\lambda)$. Substituting into $\xi^2 + \xi = \lambda^3 - \lambda$ and clearing denominators gives a degree-5 polynomial in λ that factors as

$$(\lambda - 1)(\lambda + 1)(16\lambda^3 - 9\lambda^2 + 1) = 0.$$

The roots $\lambda = \pm 1$ give $y = 0, 1$. Eliminating λ between $16\lambda^3 - 9\lambda^2 + 1 = 0$ and

$$4\lambda y - (4\lambda^3 - 3\lambda^2 - 2\lambda + 1) = 0$$

gives

$$\mathrm{Res}_\lambda\big(16\lambda^3 - 9\lambda^2 + 1,\ 4\lambda y - (4\lambda^3 - 3\lambda^2 - 2\lambda + 1)\big) = -64\, c(y).$$

The discriminant identity is the quartic reformulation. □

2.2 The cubic field

Proposition 2.3. *Let β be a root of $c(y)$, and let $F = \mathbb{Q}(\beta)$. Then:*

(i) $\mathrm{Disc}(c) = -3^9 \cdot 31^2 \cdot 37$.

(ii) Let α be a root of

$$16x^3 - 9x^2 + 1,$$

chosen so that

$$\beta = \frac{4\alpha^3 - 3\alpha^2 - 2\alpha + 1}{4\alpha}.$$

If $u := 16\alpha$, then $F = \mathbb{Q}(u)$, the element u satisfies

$$f_u(u) = 0, \qquad f_u(t) = t^3 - 9t^2 + 256,$$

and

$$\theta := 256\beta = -3u^2 + 24u - 128.$$

(iii) The field discriminant of F is

$$\mathrm{Disc}(F/\mathbb{Q}) = -999 = -3^3 \cdot 37.$$

(iv) The element

$$\omega := \frac{u^2 + 7u + 16}{32}$$

is integral, and

$$\mathcal{O}_F = \mathbb{Z} \cdot 1 \oplus \mathbb{Z}u \oplus \mathbb{Z}\omega.$$

(v) The integral model $\theta = 256\beta$ satisfies the monic polynomial

$$g(\theta) = \theta^3 + 411\theta^2 + 42240\theta + 2097152,$$

and the index of the monogenic order in $\mathcal{O}_F$ is

$$[\mathcal{O}_F : \mathbb{Z}[\theta]] = 2^8 \cdot 3^3 \cdot 31 = 214272.$$

(vi) The quadratic resolvent field of the splitting field of $c(y)$ is

$$K_\Delta = \mathbb{Q}(\sqrt{-111}).$$

Proof. Part (i) is a direct computation from (2).

For (ii), the nontrivial ramification points satisfy

$$16\lambda^3 - 9\lambda^2 + 1 = 0, \qquad y = \frac{4\lambda^3 - 3\lambda^2 - 2\lambda + 1}{4\lambda}.$$

Choosing a compatible pair (α, β) gives $F = \mathbb{Q}(\beta) = \mathbb{Q}(\alpha)$. The scaling $u = 16\alpha$ yields $f_u(u) = 0$, and substituting back gives

$$\theta = 256\beta = u^2 - 12u - 128 + \frac{1024}{u}.$$

Using $u^3 = 9u^2 - 256$, hence $1024/u = -4u^2 + 36u$, this simplifies to

$$\theta = -3u^2 + 24u - 128.$$

For (iii)–(iv), one checks that $\omega = (u^2 + 7u + 16)/32$ satisfies

$$\omega^3 - 6\omega^2 + 15\omega - 2 = 0,$$

so ω is integral. Set

$$A := \mathbb{Z} \oplus \mathbb{Z}u \oplus \mathbb{Z}\omega.$$

The change-of-basis matrix from $(1, u, u^2)$ to $(1, u, \omega)$ has determinant $1/32$, hence

$$\mathrm{Disc}(A) = \frac{\mathrm{Disc}(f_u)}{32^2}.$$

Since $\mathrm{Disc}(f_u) = -2^{10} \cdot 3^3 \cdot 37$, this gives

$$\mathrm{Disc}(A) = -3^3 \cdot 37 = -999.$$

To verify maximality of A, we check locally at 3 and 37. Since $v_{37}(\mathrm{Disc}(A)) = 1$, we have $37 \nmid [\mathcal{O}_F : A]$ from

$$\mathrm{Disc}(A) = \mathrm{Disc}(F/\mathbb{Q})\,[\mathcal{O}_F : A]^2.$$

At 3, set

$$\alpha := u - 2\omega - 1.$$

From $u^2 + 7u + 16 = 32\omega$ and $u^3 - 9u^2 + 256 = 0$,

$$\alpha^3 + 6\alpha^2 + 3\alpha + 6 = 0,$$

and

$$u = -\alpha^2 - 4\alpha + 5, \qquad \omega = 2 - \frac{\alpha^2 + 5\alpha}{2}.$$

Since $2 \in \mathbb{Z}_3^\times$, it follows that

$$A \otimes \mathbb{Z}_3 = \mathbb{Z}_3[\alpha].$$

The polynomial $x^3 + 6x^2 + 3x + 6$ is 3-Eisenstein, so $\mathbb{Z}_3[\alpha]$ is the maximal order at 3. Therefore

$$\mathcal{O}_F = A = \mathbb{Z} \oplus \mathbb{Z}u \oplus \mathbb{Z}\omega.$$

Consequently,

$$\mathrm{Disc}(F/\mathbb{Q}) = \mathrm{Disc}(A) = -999.$$

Note for later use that 3 is totally ramified in F (by the Eisenstein property) and 37 is tamely ramified ($v_{37}(\mathrm{Disc}(F)) = 1$).

For (v), $\theta = 256\beta$ satisfies $g(\theta) = 0$, and

$$\mathrm{Disc}(g) = 256^2\ \mathrm{Disc}(c) = -2^{16} \cdot 3^9 \cdot 31^2 \cdot 37.$$

Since $\theta \in \mathbb{Z}[u]$, comparing discriminants of the monogenic orders gives

$$\mathrm{Disc}(\mathbb{Z}[u]) = \mathrm{Disc}(f_u) = -2^{10} \cdot 3^3 \cdot 37$$

and therefore

$$[\mathbb{Z}[u] : \mathbb{Z}[\theta]] = \sqrt{\frac{\mathrm{Disc}(g)}{\mathrm{Disc}(f_u)}} = 2^3 \cdot 3^3 \cdot 31 = 6696.$$

Since $\omega = (u^2+7u+16)/32$, expressing $(1, u, \omega)$ in the basis $(1, u, u^2)$ gives a lower-triangular matrix with diagonal entries $(1, 1, 1/32)$; its determinant is $1/32$, whence $[\mathcal{O}_F : \mathbb{Z}[u]] = 32$. Therefore

$$[\mathcal{O}_F : \mathbb{Z}[\theta]] = [\mathcal{O}_F : \mathbb{Z}[u]]\,[\mathbb{Z}[u] : \mathbb{Z}[\theta]] = 32 \cdot 6696 = 2^8 \cdot 3^3 \cdot 31.$$

Finally, $c(y)$ is irreducible with nonsquare discriminant, so its splitting field has Galois group S_3, and its unique quadratic subfield is $\mathbb{Q}(\sqrt{\mathrm{Disc}(F)}) = \mathbb{Q}(\sqrt{-999}) = \mathbb{Q}(\sqrt{-111})$. □

Remark 2.4. The field F is recorded in the LMFDB [11] as `3.1.999.1`. Its class number is 1, so the ring of integers $\mathcal{O}_F$ is a principal ideal domain. The index

$$[\mathcal{O}_F : \mathbb{Z}[\theta]] = 2^8 \cdot 3^3 \cdot 31$$

shows that the natural generator $\theta = 256\beta$ is far from integral; the discrepancy comes from the leading coefficient $256 = 2^8$, the ramified prime 3, and the additional prime 31.

3 The regular S_4-closure

We now determine the Galois closure of the quartic cover and identify the quotient curves that will enter the Jacobian decomposition.

3.1 Monodromy and genus

Proposition 3.1. *The quartic $\Pi(\lambda, y)$ has Galois group*

$$\mathrm{Gal}(\Pi/\mathbb{Q}(y)) \cong S_4.$$

Proof. Since the discriminant $-y(y-1)c(y)$ is not a square in $\mathbb{Q}(y)$, the Galois group is not contained in A_4. Specializing at $y = 2$ gives

$$\Pi(\lambda, 2) = \lambda^4 - \lambda^3 - 5\lambda^2 + \lambda + 6.$$

Modulo 5 this is irreducible (a 4-cycle), and modulo 3 it factors as a linear times an irreducible cubic (a 3-cycle). A transitive subgroup of S_4 containing both cycle types is all of S_4. □

Theorem 3.2. *Let X be the smooth projective curve with function field equal to the Galois closure of $\mathbb{Q}(E)/\mathbb{Q}(y)$. Then $X \to \mathbb{P}^1_y$ is a regular S_4-cover of degree 24, branched with inertia orders*

$$4, 2, 2, 2, 2, 2.$$

In particular,

$$g(X) = 16.$$

Proof. The cover $E \to \mathbb{P}^1_y$ has local monodromy of order 4 at infinity and five transpositions at finite branch points (Proposition 2.2). Since the Galois group is S_4 (Proposition 3.1), the regular closure has degree 24. The inertia groups in the regular cover are generated by the corresponding monodromy elements.

The Hurwitz formula gives

$$2g(X) - 2 = 24\left(-2 + \left(1 - \frac{1}{4}\right) + 5\left(1 - \frac{1}{2}\right)\right) = 30,$$

hence $g(X) = 16$. □

3.2 Resolvent quotients

The cubic resolvent of (1), built from the three pairings $z_{ij} = \lambda_i\lambda_j + \lambda_k\lambda_l$ of the four roots [2], is

$$R(z, y) = z^3 + (2y + 1)z^2 - (4y^2 + 4y + 1)z - (8y^3 + 13y^2 + 5y + 1). \tag{5}$$

Proposition 3.3. *The discriminant of $R(z, y)$ with respect to z equals*

$$\mathrm{Disc}_z\, R(z, y) = -y(y - 1)c(y).$$

The splitting field of R over $\mathbb{Q}(y)$ is the fixed field of the Klein four subgroup $V_4 \subset S_4$.

Proof. A direct computation gives

$$\mathrm{Disc}_z\, R(z, y) = -y(y - 1)\big(256y^3 + 411y^2 + 165y + 32\big).$$

The fixed field statement is classical [2]. □

Theorem 3.4. *Let*

$$X_A := X/A_4, \qquad X_V := X/V_4.$$

Then:

(i) X_A is the genus-2 curve

$$X_A : \quad w^2 = -y(y - 1)c(y).$$

(ii) X_V is the genus-4 curve with function field

$$\mathbb{Q}(X_V) = \mathbb{Q}(y, z, \sqrt{-y(y - 1)c(y)}), \qquad R(z, y) = 0.$$

(iii) The natural map $X_V \to X_A$ is an étale cyclic cubic cover.

Proof. The A_4-quotient is the unique quadratic subextension, generated by the square root of the discriminant:

$$\mathbb{Q}(X_A) = \mathbb{Q}\big(y, \sqrt{-y(y - 1)c(y)}\big).$$

The right-hand side is squarefree of degree 5, giving genus 2. The V_4-quotient is the resolvent splitting field (Proposition 3.3).

At each finite branch point the inertia is a transposition, meeting both A_4 and V_4 trivially. At infinity the inertia is generated by a 4-cycle σ, and

$$\langle\sigma\rangle \cap A_4 = \langle\sigma^2\rangle = \langle\sigma\rangle \cap V_4.$$

The relative inertia in $A_4/V_4 \cong C_3$ is therefore trivial everywhere, so $X_V \to X_A$ is étale. Hurwitz gives

$$2g(X_V) - 2 = 3(2g(X_A) - 2) = 6,$$

hence $g(X_V) = 4$. □

Corollary 3.5. *The connected norm kernel*

$$P := \mathrm{Prym}(X_V/X_A)$$

is an abelian surface. Moreover,

$$\mathbb{Q}(\zeta_3) \hookrightarrow \mathrm{End}^0(P), \qquad \mathrm{Jac}(X_V) \sim_{\mathbb{Q}} \mathrm{Jac}(X_A) \times P.$$

Proof. The cover $X_V \to X_A$ is étale of degree 3 with $g(X_V) = 4$, so $\dim P = 4 - 2 = 2$. The standard Prym decomposition for a cyclic triple cover gives

$$\mathrm{Jac}(X_V) \sim_{\mathbb{Q}} \mathrm{Jac}(X_A) \times P$$

and the deck action furnishes $\mathbb{Q}(\zeta_3) \hookrightarrow \mathrm{End}^0(P)$ [4, Section 12.9]. □

4 The Jacobian

Our goal is to decompose $\mathrm{Jac}(X)$ up to $\mathbb{Q}$-isogeny by exploiting the S_4-module structure on $H^0(X, \Omega^1_X)$. We first work out this structure and then identify each isotypic factor with a concrete abelian variety.

4.1 Differentials

Theorem 4.1. *Let V_2 be the two-dimensional irreducible representation of S_4, V_3 the standard three-dimensional representation, and $V_3^- := V_3 \otimes \mathrm{sgn}$. Then*

$$H^0(X, \Omega^1_X) \cong \mathrm{sgn}^{\oplus 2} \oplus V_2 \oplus V_3 \oplus (V_3^-)^{\oplus 3}.$$

Accordingly there are rational isotypic factors $A_{\mathrm{sgn}}, A_{V_2}, A_{V_3}$, and $A_{V_3^-}$ such that

$$\mathrm{Jac}(X) \sim_{\mathbb{Q}} A_{\mathrm{sgn}} \times A_{V_2} \times A_{V_3} \times A_{V_3^-},$$

with dimensions

$$\dim A_{\mathrm{sgn}} = 2, \qquad \dim A_{V_2} = 2, \qquad \dim A_{V_3} = 3, \qquad \dim A_{V_3^-} = 9.$$

Proof. Since $X/S_4 \cong \mathbb{P}^1$, there is no trivial summand. Write

$$H^0(X, \Omega^1_X) \cong m_{\mathrm{sgn}} \cdot \mathrm{sgn} \oplus m_2 \cdot V_2 \oplus m_3 \cdot V_3 \oplus m_3^- \cdot V_3^-.$$

For any subgroup $H \subset S_4$, the H-invariants have dimension $g(X/H)$.

From Theorem 3.4, $g(X/A_4) = 2$ and $g(X/V_4) = 4$. If $S_3 \subset S_4$ is a point stabilizer, then $X/S_3 \cong E$, so $g(X/S_3) = 1$. Let $C = \langle (1234) \rangle$ and $D = N_{S_4}(C) \cong D_4$. Character averages give

	A_4	V_4	S_3	C	D
sgn	1	1	0	0	0
V_2	0	2	0	1	1
V_3	0	0	1	0	0
V_3^-	0	0	0	1	0

for the invariant dimensions of each irreducible constituent.

Reading off the A_4- and V_4-columns gives

$$m_{\mathrm{sgn}} = 2, \qquad m_{\mathrm{sgn}} + 2m_2 = 4,$$

hence $m_2 = 1$, and the S_3-column gives $m_3 = 1$. The genus constraint $g(X) = 16$ then forces

$$16 = m_{\mathrm{sgn}} + 2m_2 + 3m_3 + 3m_3^-,$$

so $m_3^- = 3$. The Jacobian decomposition follows from the rational isotypic decomposition. □

4.2 The dihedral quotient

Proposition 4.2. *Let $C = \langle (1234) \rangle \subset S_4$, let $D = N_{S_4}(C) \cong D_4$, and set*

$$Y := X/C, \qquad E_{\mathrm{res}} := X/D.$$

Then:

(i) The function field of E_{res} is

$$\mathbb{Q}(E_{\mathrm{res}}) = \mathbb{Q}(y, z), \qquad R(z, y) = 0,$$

where R is the cubic resolvent (5).

(ii) The smooth projective model of $R(z, y) = 0$ is birational over $\mathbb{Q}$ to the elliptic curve

$$\mathcal{E}_{\mathrm{res}} : \quad w_{\mathrm{res}}^2 = x_{\mathrm{res}}^3 - 16x_{\mathrm{res}}^2 - 64x_{\mathrm{res}} + 1040.$$

In particular,

$$g(E_{\mathrm{res}}) = 1, \qquad \Delta(\mathcal{E}_{\mathrm{res}}) = 2^{12} \cdot 31^2 \cdot 37, \qquad j(\mathcal{E}_{\mathrm{res}}) = \frac{2^{18} \cdot 7^3}{31^2 \cdot 37}.$$

(iii) $g(Y) = 4$, and the natural map $Y \to E_{\mathrm{res}}$ has degree 2 and is branched at exactly six points.

Proof. Since $V_4 \triangleleft D$ and D/V_4 has order 2, the quotient $E_{\mathrm{res}} = X/D$ sits inside the resolvent field $\mathbb{Q}(X_V)/\mathbb{Q}(y)$. Under the identification $S_4/V_4 \cong S_3$, the image of D stabilizes one of the three pairings of the four quartic roots, so the fixed field is generated by a single resolvent root z:

$$\mathbb{Q}(E_{\mathrm{res}}) = \mathbb{Q}(y, z), \qquad R(z, y) = 0.$$

The affine model $R(z, y) = 0$ passes through $(y, z) = (0, 1)$. Projecting from this point, set

$$u = \frac{z - 1}{y}.$$

Substituting $z = 1 + uy$ into $R(z, y) = 0$ gives

$$(u^3 + 2u^2 - 4u - 8)y^2 + (4u^2 - 17)y + (4u - 7) = 0.$$

Set

$$v = 2(u^3 + 2u^2 - 4u - 8)y + (4u^2 - 17).$$

Then the quadratic discriminant identity becomes

$$v^2 = -4u^3 - 16u^2 + 16u + 65.$$

After the change of variables

$$x_{\mathrm{res}} = -4u, \qquad w_{\mathrm{res}} = 4v,$$

we obtain the Weierstrass model

$$w_{\mathrm{res}}^2 = x_{\mathrm{res}}^3 - 16x_{\mathrm{res}}^2 - 64x_{\mathrm{res}} + 1040.$$

The inverse rational map is recovered from

$$u = -\frac{x_{\mathrm{res}}}{4}, \qquad v = \frac{w_{\mathrm{res}}}{4}, \qquad y = \frac{v - (4u^2 - 17)}{2(u^3 + 2u^2 - 4u - 8)}, \qquad z = 1 + uy.$$

The normalization of $R(z, y) = 0$ is therefore an elliptic curve, and the discriminant and j-invariant are read off from the integral model.

From the character table in Theorem 4.1, the invariant dimensions for C and D are

$$g(Y) = \dim H^0(X, \Omega_X^1)^C = 1 + 3 = 4, \qquad g(E_{\mathrm{res}}) = \dim H^0(X, \Omega_X^1)^D = 1.$$

Since $[D : C] = 2$, the map $Y \to E_{\mathrm{res}}$ has degree 2. Hurwitz gives

$$2g(Y) - 2 = 2(2g(E_{\mathrm{res}}) - 2) + r = r,$$

so the total ramification degree is $r = 6$. □

Proposition 4.3. *The minimal discriminant and conductor of $\mathcal{E}_{\mathrm{res}}$ are*

$$\Delta_{\min}(\mathcal{E}_{\mathrm{res}}) = 31^2 \cdot 37, \qquad N(\mathcal{E}_{\mathrm{res}}) = 31 \cdot 37 = 1147.$$

In particular, $\mathcal{E}_{\mathrm{res}}$ has multiplicative reduction at both 31 *and* 37*.*

Proof. A short computation gives $c_4 = 2^{10} \cdot 7$, $c_6 = -2^9 \cdot 23 \cdot 29$, and $\Delta = 2^{12} \cdot 31^2 \cdot 37$. Since $v_2(c_4) = 10 \geq 4$, $v_2(c_6) = 9 \geq 6$, and $v_2(\Delta) = 12 \geq 12$, one Tate reduction step with $u = 2$ yields the minimal model with

$$c_4' = c_4/2^4 = 448 = 2^6 \cdot 7, \qquad c_6' = c_6/2^6 = -5336 = -2^3 \cdot 23 \cdot 29, \qquad \Delta' = \Delta/2^{12} = 31^2 \cdot 37.$$

Since $v_2(c_6') = 3 < 6$, no further reduction at 2 is possible; at all odd primes, $v_p(c_4') < 4$ or $v_p(\Delta') = 0$, so the model is globally minimal. Now $v_{31}(c_4') = 0$ and $v_{31}(\Delta') = 2 > 0$, so the reduction at 31 is multiplicative (conductor exponent 1). Similarly $v_{37}(c_4') = 0$ and $v_{37}(\Delta') = 1 > 0$, giving multiplicative reduction at 37. □

Proposition 4.4. *Let G be a finite group acting on a smooth projective curve X over $\mathbb{Q}$, and let W be an absolutely irreducible rational representation of G such that $\mathrm{End}_G(W) = \mathbb{Q}$. Write the W-isotypic component of $H^0(X, \Omega^1_X)$ as*

$$H^0(X, \Omega^1_X)_W \cong W \otimes M_W,$$

where M_W is the multiplicity space. If $A_W \subset \mathrm{Jac}(X)$ denotes the corresponding rational isotypic factor, then there exists an abelian variety B_W, unique up to $\mathbb{Q}$-isogeny, such that

$$H^0(B_W, \Omega^1_{B_W}) \cong M_W, \qquad A_W \sim_{\mathbb{Q}} B_W^{\dim W}.$$

In particular, whenever M_W is realized by the holomorphic differentials of a quotient or Prym variety B, one has

$$A_W \sim_{\mathbb{Q}} B^{\dim W}.$$

Proof. We recall the standard group-algebra argument underlying [1]; see also [4, Section 13.4].

Let $e_W \in \mathbb{Q}[G]$ be the central idempotent attached to W. Since $\mathrm{End}_G(W) = \mathbb{Q}$, the corresponding Wedderburn block is

$$\mathbb{Q}[G]e_W \cong M_d(\mathbb{Q}), \qquad d = \dim W.$$

Choose a primitive idempotent $\varepsilon \in M_d(\mathbb{Q})$, and define

$$B_W := \varepsilon \cdot A_W.$$

The central idempotent e_W is a sum of d pairwise orthogonal conjugates of ε, so

$$A_W \sim_{\mathbb{Q}} B_W^d.$$

On differentials, $H^0(A_W, \Omega^1_{A_W}) \cong W \otimes M_W$, and ε picks out a single matrix-column factor:

$$H^0(B_W, \Omega^1_{B_W}) \cong M_W.$$

Any other abelian variety B realizing the same multiplicity space defines the same rational Hodge substructure of $H^1(X)$, hence is $\mathbb{Q}$-isogenous to B_W. □

Remark 4.5. Every complex irreducible representation of S_4 is realizable over $\mathbb{Q}$ (Schur index 1), so the hypothesis $\mathrm{End}_G(W) = \mathbb{Q}$ of Proposition 4.4 is automatically fulfilled for each of the four isotypic constituents.

Proposition 4.6. *Let W_{sgn} and W_{V_2} be the sign and V_2-isotypic summands of $H^0(X, \Omega^1_X)$. Then*

$$H^0(X, \Omega^1_X)^{A_4} = W_{\mathrm{sgn}}, \qquad H^0(X, \Omega^1_X)^{V_4} = W_{\mathrm{sgn}} \oplus W_{V_2}.$$

Moreover, V_4 acts trivially on V_2, and the quotient group $A_4/V_4 \cong C_3$ acts trivially on W_{sgn} and through the unique nontrivial irreducible rational representation on W_{V_2}. Consequently,

$$A_{\mathrm{sgn}} \sim_{\mathbb{Q}} \mathrm{Jac}(X_A), \qquad H^0(P, \Omega^1_P) \cong W_{V_2}, \qquad P \sim_{\mathbb{Q}} A_{V_2}.$$

Proof. From the character table computed in Theorem 4.1, the only constituent with nonzero A_4-invariants is sgn, contributing one invariant line per copy. Hence

$$H^0(X, \Omega^1_X)^{A_4} = W_{\mathrm{sgn}}.$$

For V_4, the same table shows that the only constituents with nonzero invariants are sgn and V_2, with invariant dimensions 1 and 2, respectively. Therefore

$$H^0(X, \Omega^1_X)^{V_4} = W_{\mathrm{sgn}} \oplus W_{V_2}.$$

The character of V_2 evaluates to 2 on every element of V_4, so V_4 acts trivially on W_{V_2}. Since V_2 has no A_4-invariants, the quotient $A_4/V_4 \cong C_3$ acts without fixed vectors, forcing W_{V_2} to be the unique nontrivial irreducible rational C_3-module. The sign representation is trivial on A_4, so the action on W_{sgn} is trivial.

Pullback identifies $H^0(X_A, \Omega^1_{X_A})$ with $H^0(X, \Omega^1_X)^{A_4} = W_{\mathrm{sgn}}$. Since sgn is one-dimensional, Proposition 4.4 gives

$$A_{\mathrm{sgn}} \sim_{\mathbb{Q}} \mathrm{Jac}(X_A).$$

Likewise,

$$H^0(X_V, \Omega^1_{X_V}) = H^0(X, \Omega^1_X)^{V_4} = W_{\mathrm{sgn}} \oplus W_{V_2}.$$

By Corollary 3.5,

$$H^0(X_V, \Omega^1_{X_V}) \cong \pi^* H^0(X_A, \Omega^1_{X_A}) \oplus H^0(P, \Omega^1_P),$$

where $\pi : X_V \to X_A$ is the natural cover. Comparing the two decompositions shows that

$$H^0(P, \Omega^1_P) \cong W_{V_2}.$$

It follows that P and A_{V_2} define the same rational Hodge substructure of $H^1(X)$, so $P \sim_{\mathbb{Q}} A_{V_2}$. □

4.3 The factorization

Theorem 4.7. *Let $E = X/S_3$ be the conductor-37 elliptic quotient from Theorem 2.1, let $E_{\mathrm{res}} = X/D_4$ be the resolvent elliptic quotient from Proposition 4.2, and let*

$$Q := \mathrm{Prym}(Y/E_{\mathrm{res}}), \qquad Y = X/C_4.$$

Then one has

$$A_{\mathrm{sgn}} \sim_{\mathbb{Q}} \mathrm{Jac}(X_A), \qquad P \sim_{\mathbb{Q}} E^2_{\mathrm{res}}, \qquad A_{V_3} \sim_{\mathbb{Q}} E^3, \qquad A_{V_3^-} \sim_{\mathbb{Q}} Q^3.$$

Consequently,

$$\mathrm{Jac}(X) \sim_{\mathbb{Q}} \mathrm{Jac}(X_A) \times E^2_{\mathrm{res}} \times E^3 \times Q^3.$$

Proof. Proposition 4.6 identifies the sign constituent:

$$A_{\mathrm{sgn}} \sim_{\mathbb{Q}} \mathrm{Jac}(X_A).$$

We identify the remaining three factors in turn.

For V_3, let $H \subset S_4$ be a point stabilizer, so that $X/H \cong E$. The invariant table shows that V_3 is the unique constituent with nonzero H-invariants, giving

$$H^0(E, \Omega^1_E) \cong M_{V_3},$$

where M_{V_3} is the multiplicity space of V_3. Proposition 4.4 gives

$$A_{V_3} \sim_{\mathbb{Q}} E^3.$$

For V_2, we already know $P \sim_{\mathbb{Q}} A_{V_2}$ from Proposition 4.6. The quotient $E_{\mathrm{res}} = X/D$ satisfies

$$H^0(E_{\mathrm{res}}, \Omega^1_{E_{\mathrm{res}}}) = H^0(X, \Omega^1_X)^D,$$

and the D-invariant column shows that this is the multiplicity space M_{V_2}. Proposition 4.4 then gives

$$A_{V_2} \sim_{\mathbb{Q}} E^2_{\mathrm{res}}.$$

Together with Proposition 4.6, this yields

$$P \sim_{\mathbb{Q}} E^2_{\mathrm{res}}.$$

The most delicate factor is V_3^-. The degree-2 map $Y \to E_{\mathrm{res}}$ has a deck involution $\iota \in D/C$, and the pullback decomposition gives

$$H^0(Y, \Omega^1_Y) \cong \pi^* H^0(E_{\mathrm{res}}, \Omega^1_{E_{\mathrm{res}}}) \oplus H^0(Q, \Omega^1_Q),$$

where the first summand is the ι-invariant part and the second is the ι-anti-invariant part. Now

$$H^0(Y, \Omega^1_Y) = H^0(X, \Omega^1_X)^C.$$

The character table gives $\dim H^0(X, \Omega^1_X)^C = 4$ (one line from V_2, one from each of the three copies of V_3^-) and $\dim H^0(X, \Omega^1_X)^D = 1$ (from V_2 alone). Since V_3^- has vanishing character average over D, its C-invariant lines all lie in the ι-anti-invariant eigenspace, so

$$H^0(Q, \Omega^1_Q) \cong M_{V_3^-},$$

and $\dim Q = 3$. Proposition 4.4 (applicable by Remark 4.5) now gives

$$A_{V_3^-} \sim_{\mathbb{Q}} Q^3.$$

Assembling the four identifications gives

$$\mathrm{Jac}(X) \sim_{\mathbb{Q}} \mathrm{Jac}(X_A) \times E^2_{\mathrm{res}} \times E^3 \times Q^3.$$

□

Remark 4.8. The isogeny $P \sim_{\mathbb{Q}} E_{\mathrm{res}}^2$ is induced by the same rational correspondence on X that realizes the V_2-isotypic projector in $\mathbb{Q}[S_4]$. On the one hand, if σ generates the deck group of the étale cyclic triple cover $X_V \to X_A$, then the Prym projector $1-\sigma$ cuts out the anti-invariant part of $\mathrm{Jac}(X_V)$, namely P. On the other hand, the quotient map $X \to X/D_4 = E_{\mathrm{res}}$ identifies the same multiplicity space with $H^0(E_{\mathrm{res}}, \Omega^1_{E_{\mathrm{res}}})$. The resulting pullback and norm correspondences induce homomorphisms

$$P \longrightarrow E_{\mathrm{res}}^2, \qquad E_{\mathrm{res}}^2 \longrightarrow P$$

which are inverse up to finite kernel. This is the concrete V_2-block of the general functorial decomposition formalism for Jacobians of Galois covers [12]. We do not compute the minimal degree of this isogeny here.

Corollary 4.9. *There is a corresponding factorization of first cohomology and of Hasse–Weil L-functions:*

$$H^1(X) \sim H^1(X_A) \oplus H^1(E_{\mathrm{res}})^{\oplus 2} \oplus H^1(E)^{\oplus 3} \oplus H^1(Q)^{\oplus 3},$$

hence

$$L(H^1(X), s) = L(H^1(X_A), s)\, L(E_{\mathrm{res}}, s)^2\, L(E, s)^3\, L(H^1(Q), s)^3.$$

Proof. This follows at once from Theorem 4.7 by passing to ℓ-adic representations and taking Euler products. □

Proposition 4.10. *The Prym surface $P = \mathrm{Prym}(X_V/X_A)$ carries a polarization of type $(1,3)$ and satisfies*

$$\mathrm{End}^0(P) \cong M_2(\mathbb{Q}), \qquad \rho(P_{\overline{\mathbb{Q}}}) = 3.$$

In particular, P is not of CM type.

Proof. The Prym polarization for an étale cyclic triple cover of a genus-2 curve has type $(1,3)$ [4, Section 12.9]. Since

$$P \sim_{\mathbb{Q}} E_{\mathrm{res}}^2.$$

by Theorem 4.7, and the j-invariant $j(E_{\mathrm{res}}) = 2^{18} \cdot 7^3/(31^2 \cdot 37)$ is not an algebraic integer, the curve E_{res} has no complex multiplication [9, Ch. II]. It follows that

$$\mathrm{End}^0(P) \cong \mathrm{End}^0(E_{\mathrm{res}}^2) \cong M_2(\mathbb{Q}).$$

Isogeny induces an isomorphism on Néron–Severi groups after tensoring with $\mathbb{Q}$, so

$$\rho(P_{\overline{\mathbb{Q}}}) = \rho((E_{\mathrm{res}}^2)_{\overline{\mathbb{Q}}}).$$

For the square of a non-CM elliptic curve the geometric Picard number is 3, generated by the two coordinate classes and the diagonal. Since $\mathrm{End}^0(P) \cong M_2(\mathbb{Q})$ is not a commutative semisimple algebra of dimension $2\dim P$, the surface P is not of CM type. □

Remark 4.11. The cover $X_V \to X_A$ fits the cyclic triple-cover setting of Lange–Ortega [8]. Their analysis of non-cyclic triple coverings of genus-2 curves [13] exhibits a markedly different Prym behavior, with principally polarized Prym surfaces on the non-cyclic side. In the present cyclic example the Prym polarization has type $(1,3)$, and the additional S_4-symmetry forces the stronger splitting

$$P \sim_{\mathbb{Q}} E_{\mathrm{res}}^2.$$

5 Ray classes and weight one

We turn to the class-field-theoretic description of the splitting field of the branch cubic. Let $F = \mathbb{Q}(\beta)$ be the cubic field generated by a root β of $c(y)$, let L be the splitting field, and let

$$K = \mathbb{Q}(\sqrt{-111})$$

be the quadratic resolvent field.

Proposition 5.1. *The ideal class group of $K = \mathbb{Q}(\sqrt{-111})$ is cyclic of order 8:*

$$\mathrm{Cl}(K) \cong C_8.$$

Proof. Let

$$\tau := \frac{1+\sqrt{-111}}{2},$$

so $\mathcal{O}_K = \mathbb{Z}[\tau]$ and $\tau^2 - \tau + 28 = 0$. Since $\mathrm{Disc}(K) = -111$, the primitive reduced positive definite binary quadratic forms of discriminant -111 are

$$(1,1,28),\ (2,\pm 1,14),\ (3,3,10),\ (4,\pm 1,7),\ (5,\pm 3,6).$$

There are eight such forms, so $h_K = 8$.

To exhibit a generator, consider the ideal

$$\mathfrak{a} := (2, \tau - 1) \subset \mathcal{O}_K.$$

A direct ideal multiplication gives

$$\mathfrak{a}^2 = (4, \tau+3), \qquad \mathfrak{a}^4 = (16, \tau+11), \qquad \mathfrak{a}^8 = (1+3\tau).$$

so $[\mathfrak{a}]^8 = 1$ in $\mathrm{Cl}(K)$. To verify that the order is exactly 8, note that the norm form on $\mathcal{O}_K$ is

$$N_{K/\mathbb{Q}}(x + y\tau) = x^2 + xy + 28y^2.$$

If $\mathfrak{a}^4$ were principal, it would be generated by an element of norm 16. Solving

$$x^2 + xy + 28y^2 = 16$$

gives only $(x, y) = (\pm 4, 0)$, so the only principal ideal of norm 16 is (4). But $\mathfrak{a}^4 = (16, \tau+11) \neq (4)$, so $[\mathfrak{a}]$ has order exactly 8 and generates $\mathrm{Cl}(K)$. □

Theorem 5.2. *Let $\mathfrak{p}_3$ be the unique prime of K above 3. Then:*

(i) L/K is cyclic of degree 3.

(ii) The Artin conductor of L/K is

$$\mathfrak{f}_{L/K} = \mathfrak{p}_3^2.$$

(iii) The relative discriminant is

$$\mathfrak{d}_{L/K} = \mathfrak{p}_3^4.$$

(iv) The absolute discriminant of L is

$$\mathrm{Disc}(L/\mathbb{Q}) = 3^7 \cdot 37^3.$$

Proof. Since $c(y)$ is irreducible with nonsquare discriminant (Proposition 2.3), the Galois group of $L/\mathbb{Q}$ is S_3, and its unique index-2 subgroup fixes K. Hence L/K is cyclic of degree 3.

Let ρ denote the standard two-dimensional Artin representation of S_3. The factorization

$$\zeta_F(s) = \zeta(s)L(\rho, s)$$

shows that the Artin conductor of ρ is

$$\mathfrak{f}(\rho) = |\operatorname{Disc}(F/\mathbb{Q})| = 999.$$

On the other hand, if χ is a nontrivial character of $\operatorname{Gal}(L/K)$, then

$$\rho \cong \operatorname{Ind}_{G_K}^{G_{\mathbb{Q}}} \chi.$$

The conductor formula for induction gives [5, Ch. VII]

$$\mathfrak{f}(\rho) = |\operatorname{Disc}(K/\mathbb{Q})|\, N_{K/\mathbb{Q}}(\mathfrak{f}_\chi).$$

Since $|\operatorname{Disc}(K/\mathbb{Q})| = 111$, we obtain

$$N_{K/\mathbb{Q}}(\mathfrak{f}_\chi) = \frac{999}{111} = 9.$$

Since 3 ramifies in K, there is a unique prime $\mathfrak{p}_3$ above 3, and $\mathfrak{p}_3^2$ is the only ideal of norm 9. Therefore

$$\mathfrak{f}_\chi = \mathfrak{p}_3^2.$$

The two nontrivial characters of a cyclic cubic extension share the same conductor, so

$$\mathfrak{f}_{L/K} = \mathfrak{p}_3^2$$

and the conductor–discriminant formula [5, Ch. VII] gives $\mathfrak{d}_{L/K} = \mathfrak{p}_3^4$. Taking norms,

$$N_{K/\mathbb{Q}}(\mathfrak{d}_{L/K}) = N(\mathfrak{p}_3)^4 = 3^4.$$

Therefore

$$\operatorname{Disc}(L/\mathbb{Q}) = \operatorname{Disc}(K/\mathbb{Q})^3\, N(\mathfrak{d}_{L/K}) = 111^3 \cdot 3^4 = 3^7 \cdot 37^3.$$

□

Remark 5.3. The two discriminants appearing here refer to different fields and should not be conflated. The cubic field $F = \mathbb{Q}(\beta)$ is the non-Galois cubic subfield of the S_3-extension $L/\mathbb{Q}$, and

$$\operatorname{Disc}(F/\mathbb{Q}) = -999 = -3^3 \cdot 37$$

by Proposition 2.3. By contrast, L is the full splitting field, so its absolute discriminant is obtained from the quadratic resolvent field $K = \mathbb{Q}(\sqrt{-111})$ and the relative discriminant of the cubic extension L/K:

$$\operatorname{Disc}(L/\mathbb{Q}) = \operatorname{Disc}(K/\mathbb{Q})^3\, N_{K/\mathbb{Q}}(\mathfrak{d}_{L/K}) = 111^3 \cdot 3^4 = 3^7 \cdot 37^3.$$

Thus items (ii)–(iv) are mutually consistent: $\mathfrak{f}_{L/K} = \mathfrak{p}_3^2$ implies $\mathfrak{d}_{L/K} = \mathfrak{p}_3^4$, and after taking norms this yields the displayed discriminant of L, not the discriminant of the cubic subfield F.

Theorem 5.4. *Let H_K be the Hilbert class field of $K = \mathbb{Q}(\sqrt{-111})$, and let $K_{\mathfrak{p}_3^2}$ be the ray class field of modulus $\mathfrak{p}_3^2$. Then:*

(i) The ray class group is cyclic:

$$\mathrm{Cl}_{\mathfrak{p}_3^2}(K) \cong C_{24}.$$

(ii) The ray class field is the compositum

$$K_{\mathfrak{p}_3^2} = H_K L, \qquad H_K \cap L = K.$$

(iii) The extension $K_{\mathfrak{p}_3^2}/\mathbb{Q}$ is Galois with generalized dihedral group

$$\mathrm{Gal}(K_{\mathfrak{p}_3^2}/\mathbb{Q}) \cong C_{24} \rtimes C_2,$$

where the nontrivial element of C_2 acts on C_{24} by inversion.

(iv) The splitting field L of $c(y)$ is the unique cubic subextension of $K_{\mathfrak{p}_3^2}/K$.

Proof. The standard exact sequence for the modulus $\mathfrak{p}_3^2$ reads [5, Ch. VI]

$$1 \longrightarrow \mathcal{O}_K^\times/(\mathcal{O}_K^\times \cap (1+\mathfrak{p}_3^2)) \longrightarrow (\mathcal{O}_K/\mathfrak{p}_3^2)^\times \longrightarrow \mathrm{Cl}_{\mathfrak{p}_3^2}(K) \longrightarrow \mathrm{Cl}(K) \longrightarrow 1.$$

Since K is imaginary quadratic with discriminant -111, the unit group is $\mathcal{O}_K^\times = \{\pm 1\}$, and $-1 \not\equiv 1 \pmod{\mathfrak{p}_3^2}$, so its image in $(\mathcal{O}_K/\mathfrak{p}_3^2)^\times$ has order 2. The residue field of $\mathfrak{p}_3$ has size 3, so

$$|(\mathcal{O}_K/\mathfrak{p}_3^2)^\times| = N(\mathfrak{p}_3^2)\left(1 - \frac{1}{N(\mathfrak{p}_3)}\right) = 9\left(1 - \frac{1}{3}\right) = 6.$$

Equivalently, because 3 is ramified in K, the local kernel is

$$\frac{(\mathcal{O}_K/\mathfrak{p}_3^2)^\times}{\{\pm 1\}} \cong \frac{1+\mathfrak{p}_3}{1+\mathfrak{p}_3^2} \cong \frac{\mathfrak{p}_3}{\mathfrak{p}_3^2} \cong \mathbb{F}_3,$$

so it is cyclic of order 3. Thus the kernel of $\mathrm{Cl}_{\mathfrak{p}_3^2}(K) \to \mathrm{Cl}(K)$ has order 3. Since $\mathrm{Cl}(K) \cong C_8$ by Proposition 5.1,

$$|\,\mathrm{Cl}_{\mathfrak{p}_3^2}(K)| = 3 \cdot 8 = 24.$$

Let $[\mathfrak{a}]$ be the generator of $\mathrm{Cl}(K)$ from Proposition 5.1, and let u be any nontrivial element of the local kernel. The kernel has order coprime to 8, so the sequence splits and

$$\mathrm{Cl}_{\mathfrak{p}_3^2}(K) \cong C_8 \times C_3 \cong C_{24},$$

with generators $[\mathfrak{a}]$ and u. This proves (i).

The Hilbert class field satisfies $[H_K : K] = h_K = 8$, and L/K is cyclic cubic of conductor exactly $\mathfrak{p}_3^2$ (Theorem 5.2). Since $\gcd(8,3) = 1$,

$$H_K \cap L = K.$$

Hence

$$[H_K L : K] = [H_K : K]\,[L : K] = 24 = [K_{\mathfrak{p}_3^2} : K],$$

so $K_{\mathfrak{p}_3^2} = H_K L$. This proves (ii).

The modulus $\mathfrak{p}_3^2$ is stable under complex conjugation, so $K_{\mathfrak{p}_3^2}/\mathbb{Q}$ is Galois and conjugation acts on $\mathrm{Cl}_{\mathfrak{p}_3^2}(K)$ by inversion [2, Ch. 9], giving (iii).

For (iv), C_{24} has a unique subgroup of index 3, and L/K corresponds to this subgroup by Theorem 5.2. □

Remark 5.5. The proof gives a concrete presentation of the ray class group:

$$\mathrm{Cl}_{\mathfrak{p}_3^2}(K) = \langle [\mathfrak{a}], u \mid [\mathfrak{a}]^8 = u^3 = 1,\ [\mathfrak{a}]u = u[\mathfrak{a}] \rangle,$$

where $\mathfrak{a} = (2, \tau - 1)$ and u is any nontrivial class in $(1 + \mathfrak{p}_3)/(1 + \mathfrak{p}_3^2)$. The companion audit file records the powers of $\mathfrak{a}$ and the corresponding ray-class arithmetic used in Section 5.

Corollary 5.6. *Let ρ_c be the standard two-dimensional Artin representation attached to the splitting field $L/\mathbb{Q}$ of $c(y)$. Then*

$$N(\rho_c) = 999, \qquad \det(\rho_c) = \chi_{-111},$$

and there exists a normalized weight-1 cuspidal newform

$$f_c(q) = \sum_{n \geq 1} a_n q^n \in S_1(\Gamma_0(999), \chi_{-111})$$

whose L-function equals $L(\rho_c, s)$.

Proof. The Artin conductor is $|\mathrm{Disc}(F/\mathbb{Q})| = 999$ by Proposition 2.3, and the determinant is the quadratic character attached to $K = \mathbb{Q}(\sqrt{-111})$. Writing

$$\rho_c \cong \mathrm{Ind}_{G_K}^{G_\mathbb{Q}} \chi$$

for a finite-order character χ of $\mathrm{Gal}(L/K)$, we see that ρ_c has finite image. Equivalently, ρ_c is the nontrivial two-dimensional summand of the permutation representation of $G_\mathbb{Q}$ on the three roots of $c(y)$: since $c(y)$ is irreducible with Galois group S_3, this summand is irreducible. Its determinant is the sign character of S_3, namely χ_{-111}, so complex conjugation acts with determinant -1 and ρ_c is odd. The theorem of Deligne–Serre [14] therefore attaches a normalized weight-1 cuspidal newform of level 999 and nebentypus χ_{-111} whose L-function equals $L(\rho_c, s)$. □

Theorem 5.7. *Let $f_c(q) = \sum_{n \geq 1} a_n q^n$ be the normalized newform from Corollary 5.6. Then:*

(i) For every prime $p \nmid 2 \cdot 3 \cdot 31 \cdot 37$,

$$a_p = \#\{y \in \mathbb{F}_p :\ c(y) = 0\} - 1.$$

(ii) For every prime $p \nmid 2 \cdot 3 \cdot 37$,

$$a_p = \#\{u \in \mathbb{F}_p :\ u^3 - 9u^2 + 256 = 0\} - 1.$$

In particular, the prime 31 is the unique prime away from the Artin conductor at which the maximal-order model $u^3 - 9u^2 + 256$ still detects the Frobenius trace, while the original branch cubic $c(y)$ does not.

Proof. For p outside the exceptional set in (i), the reduction of $c(y)$ modulo p is separable, so its factorization type encodes the cycle type of Frob_p. Let ρ_{perm} be the 3-dimensional permutation representation on the roots of $c(y)$. Then

$$\rho_{\mathrm{perm}} \cong \mathbf{1} \oplus \rho_c,$$

and the trace of $\rho_{\mathrm{perm}}(\mathrm{Frob}_p)$ is the number of roots of $c(y)$ fixed by Frobenius, i.e. the number of roots of $c(y)$ in $\mathbb{F}_p$. Therefore

$$a_p = \mathrm{tr}\, \rho_c(\mathrm{Frob}_p) = \mathrm{tr}\, \rho_{\mathrm{perm}}(\mathrm{Frob}_p) - 1 = \#\mathrm{Fix}(\mathrm{Frob}_p) - 1 = \#\{y \in \mathbb{F}_p :\ c(y) = 0\} - 1.$$

The same reasoning applies to the monic model $u^3 - 9u^2 + 256$ in (ii). The exceptional sets differ by the single prime 31: the nonmonic model $c(y)$ drops degree modulo 31 (since $256 \equiv 0$), while the maximal-order model remains separable there. □

6 The genus-two Jacobian

The genus-2 factor $\mathrm{Jac}(X_A)$ carries the richest arithmetic information in the decomposition. In this section we record Frobenius data at small good primes, establish the automorphic decomposition of $L(H^1(X), s)$, obtain a Picard-number lower bound for $\mathrm{Jac}(X)$, and analyze the local reduction and Galois representations of $\mathrm{Jac}(X_A)$.

6.1 Frobenius data at small good primes

Proposition 6.1. *For the genus-two curve*

$$X_A : \quad w^2 = -y(y-1)(256y^3 + 411y^2 + 165y + 32),$$

the Frobenius characteristic polynomials at the good primes $p = 5, 13, 17$ are

$$\chi_5(T) = T^4 - T^3 + 4T^2 - 5T + 25 = (T^2 + 2T + 5)(T^2 - 3T + 5), \tag{6}$$

$$\chi_{13}(T) = T^4 - 2T^2 + 169, \tag{7}$$

$$\chi_{17}(T) = T^4 - 3T^3 + 28T^2 - 51T + 289. \tag{8}$$

In particular, $\chi_{13}(T)$ is irreducible over $\mathbb{Q}$. Moreover,

$$\chi_{13}(T) = (T^2 - 2\sqrt{7}\,T + 13)(T^2 + 2\sqrt{7}\,T + 13)$$

over $\mathbb{Q}(\sqrt{7})$, while

$$\chi_{17}(T) = \left(T^2 - \frac{3+\sqrt{33}}{2}\,T + 17\right)\left(T^2 - \frac{3-\sqrt{33}}{2}\,T + 17\right)$$

over $\mathbb{Q}(\sqrt{33})$.

More generally, for any good prime p write

$$\chi_p(T) = T^4 - s_1(p)T^3 + s_2(p)T^2 - s_1(p)pT + p^2.$$

Then $\chi_p(T)$ is reducible over $\mathbb{Q}$ if and only if

$$\Delta_p := s_1(p)^2 - 4\big(s_2(p) - 2p\big)$$

is a square in $\mathbb{Q}$. For

$$p = 13, 17, 23, 29, 43, 47, 53$$

one finds

$$\Delta_p = 112,\ 33,\ 201,\ 73,\ 304,\ 228,\ 112,$$

so $\chi_p(T)$ is irreducible over $\mathbb{Q}$ at each of these primes.

Proof. Point counting on X_A gives

$$\#X_A(\mathbb{F}_5) = 5, \qquad \#X_A(\mathbb{F}_{25}) = 33,$$

$$\#X_A(\mathbb{F}_{13}) = 14, \qquad \#X_A(\mathbb{F}_{169}) = 166,$$

$$\#X_A(\mathbb{F}_{17}) = 15, \qquad \#X_A(\mathbb{F}_{289}) = 337.$$

Hence

$$(s_1, s_2) = (1, 4) \text{ at } p = 5, \qquad (s_1, s_2) = (0, -2) \text{ at } p = 13, \qquad (s_1, s_2) = (3, 28) \text{ at } p = 17,$$

which yields (6), (7), and (8).

For a good prime p, any factorization of

$$\chi_p(T) = T^4 - s_1 T^3 + s_2 T^2 - s_1 p T + p^2$$

over $\mathbb{Q}$ must, by the rational-root test and Gauss's lemma, have the form

$$\chi_p(T) = (T^2 - uT + p)(T^2 - vT + p),$$

since the constant term is p^2 and the reciprocal symmetry forces each quadratic factor to have constant term p. Comparing coefficients gives

$$u + v = s_1, \qquad uv = s_2 - 2p.$$

Therefore $\chi_p(T)$ is reducible over $\mathbb{Q}$ if and only if the quadratic equation

$$X^2 - s_1 X + (s_2 - 2p) = 0$$

has a rational root, equivalently if and only if

$$\Delta_p = s_1^2 - 4(s_2 - 2p)$$

is a square in $\mathbb{Q}$. At $p = 5$, one has $\Delta_5 = 25$, which yields the displayed factorization. At $p = 13$ and $p = 17$, the values are $\Delta_{13} = 112$ and $\Delta_{17} = 33$, so both quartics are irreducible over $\mathbb{Q}$. The same computation from the point-count data gives

$$(s_1, s_2) = (3, -2),\ (-1, 40),\ (-4, 14),\ (6, 46),\ (-8, 94)$$

at $p = 23, 29, 43, 47, 53$, respectively, hence the additional nonsquare values $201, 73, 304, 228, 112$.

The displayed factorizations over $\mathbb{Q}(\sqrt{7})$ and $\mathbb{Q}(\sqrt{33})$ are obtained by solving

$$u + v = s_1, \qquad uv = s_2 - 2p,$$

for the quadratic traces u, v of Frobenius on the two degree-2 factors. □

Remark 6.2. These repeated irreducibility witnesses do not by themselves determine the geometric endomorphism algebra of $\mathrm{Jac}(X_A)$. They do, however, make accidental decompositions of $\mathrm{Jac}(X_A)$ into lower-dimensional isogeny factors increasingly implausible. In particular, the extra good primes up to 53 reinforce the distinction between the genus-2 Jacobian and the explicitly split Prym surface $P \sim_{\mathbb{Q}} E_{\mathrm{res}}^2$.

6.2 Automorphic structure

Theorem 6.3. *The L-function of the first cohomology of the regular S_4-closure X admits the factorization*

$$L(H^1(X), s) = L(\mathrm{Jac}(X_A), s)\, L(f_{1147}, s)^2\, L(f_{37}, s)^3\, L(H^1(Q), s)^3, \tag{9}$$

where:

(i) $f_{37} \in S_2(\Gamma_0(37))$ *is the unique normalized newform of level* 37*, attached to the elliptic curve* $E = 37\mathrm{a}1$*;*

(ii) $f_{1147} \in S_2(\Gamma_0(1147))$ *is the newform attached to the resolvent elliptic curve* E_{res}*, which has conductor* 1147 *and multiplicative reduction at both* 31 *and* 37*;*

(iii) $L(\mathrm{Jac}(X_A), s)$ *is the degree-*4 *Hasse–Weil L-function of the genus-*2 *Jacobian* $\mathrm{Jac}(X_A)$*;*

(iv) $Q = \mathrm{Prym}(Y/E_{\mathrm{res}})$ *is an abelian threefold;* $H^1(Q)$ *carries the* V_3^-*-multiplicity factor.*

Proof. Theorem 4.7 gives the Hodge-structure decomposition, and the L-function identity (9) follows by taking Euler products. Modularity of $L(E, s)$ and $L(E_{\mathrm{res}}, s)$ is the modularity theorem for elliptic curves over $\mathbb{Q}$ [15]. Proposition 4.3 identifies E_{res} as an elliptic curve of conductor 1147. The remaining assertions are definitions of the factors appearing in the decomposition. □

Corollary 6.4. *The Hasse–Weil L-function* $L(\mathrm{Jac}(X_A), s)$ *has meromorphic continuation to* $\mathbb{C}$ *and satisfies the expected functional equation.*

Proof. Every abelian surface over a totally real field is potentially modular [16]; in particular $L(\mathrm{Jac}(X_A), s)$ has meromorphic continuation and satisfies the expected functional equation. □

Remark 6.5. The two abelian surface factors of $\mathrm{Jac}(X)$ are structurally quite different. The sign factor is the Jacobian $\mathrm{Jac}(X_A)$ of a concrete genus-2 curve, whereas the V_2-factor $P \sim E_{\mathrm{res}}^2$ is explicitly split and has endomorphism algebra $M_2(\mathbb{Q})$. Thus the same branch cubic $c(y)$ produces both a Jacobian-type abelian surface and a non-simple Prym surface.

6.3 Explicit endomorphism blocks

Proposition 6.6. *The elliptic curves* E *and* E_{res} *are not* $\mathbb{Q}$*-isogenous. Moreover,*

$$\mathrm{End}^0(\mathrm{Jac}(X)_{\overline{\mathbb{Q}}}) \supseteq \mathrm{End}^0(\mathrm{Jac}(X_A)_{\overline{\mathbb{Q}}}) \times M_2(\mathbb{Q}) \times M_3(\mathbb{Q}) \times \mathrm{End}^0(Q^3_{\overline{\mathbb{Q}}}).$$

Proof. For the first assertion, $\mathbb{Q}$-isogenous elliptic curves have the same conductor. Since

$$\mathrm{cond}(E) = 37 \neq 1147 = \mathrm{cond}(E_{\mathrm{res}}),$$

the curves E and E_{res} are not $\mathbb{Q}$-isogenous.

The displayed containment is immediate from the $\mathbb{Q}$-isogeny

$$\mathrm{Jac}(X) \sim_{\mathbb{Q}} \mathrm{Jac}(X_A) \times E_{\mathrm{res}}^2 \times E^3 \times Q^3$$

of Theorem 4.7. The matrix blocks $M_2(\mathbb{Q})$ and $M_3(\mathbb{Q})$ come from the standard endomorphisms of the powers E_{res}^2 and E^3. □

Corollary 6.7. *The geometric Picard number of* $\mathrm{Jac}(X)$ *satisfies*

$$\rho(\mathrm{Jac}(X)_{\overline{\mathbb{Q}}}) \geq 16.$$

Proof. The isogeny decomposition of Theorem 4.7 yields

$$\rho(\mathrm{Jac}(X)_{\overline{\mathbb{Q}}}) \geq \rho(\mathrm{Jac}(X_A)_{\overline{\mathbb{Q}}}) + \rho(E^2_{\mathrm{res}\overline{\mathbb{Q}}}) + \rho(E^3_{\overline{\mathbb{Q}}}) + \rho(Q^3_{\overline{\mathbb{Q}}}).$$

Now $\rho(\mathrm{Jac}(X_A)_{\overline{\mathbb{Q}}}) \geq 1$. For the non-CM elliptic curves E_{res} and E, the Rosati-symmetric parts of $M_2(\mathbb{Q})$ and $M_3(\mathbb{Q})$ have dimensions 3 and 6, so

$$\rho(E_{\mathrm{res}\overline{\mathbb{Q}}}^2) = 3, \qquad \rho(E_{\overline{\mathbb{Q}}}^3) = 6.$$

Finally, Q^3 carries the obvious matrix endomorphisms from the three coordinate factors, so its Rosati-symmetric endomorphisms contain the 6-dimensional space of symmetric 3×3 rational matrices. Hence $\rho(Q_{\overline{\mathbb{Q}}}^3) \geq 6$, and therefore

$$\rho(\mathrm{Jac}(X)_{\overline{\mathbb{Q}}}) \geq 1 + 3 + 6 + 6 = 16.$$

□

6.4 The Prym threefold

Proposition 6.8. *The abelian threefold $Q = \mathrm{Prym}(Y/E_{\mathrm{res}})$ satisfies:*

(i) $\mathrm{Jac}(Y) \sim_{\mathbb{Q}} E_{\mathrm{res}} \times Q$.

(ii) The ℓ-adic Galois representation on the first cohomology of Q is

$$V_\ell(Q) \cong M_{V_3^-} \otimes \mathbb{Q}_\ell,$$

where $M_{V_3^-}$ is the multiplicity space of the anti-standard representation $V_3^- = V_3 \otimes \mathrm{sgn}$ of S_4.

(iii) The L-function of Q is a degree-6 Euler product. The factorization (9) gives

$$L(H^1(Q), s)^3 = \frac{L(H^1(X), s)}{L(\mathrm{Jac}(X_A), s)\, L(E_{\mathrm{res}}, s)^2\, L(E, s)^3},$$

so $L(H^1(Q), s)$ is determined by the L-functions of the remaining factors and the genus-16 curve X.

Proof. Part (i) is the standard Prym decomposition for the degree-2 cover $Y \to E_{\mathrm{res}}$ [3, 8].

For (ii), Theorem 4.7 identifies $H^0(Q, \Omega_Q^1)$ with $M_{V_3^-}$; the same identification holds at the ℓ-adic level. Part (iii) follows by dividing (9) by the three known factors and comparing Euler products at unramified primes. □

Proposition 6.9. *For every prime $p \nmid 2 \cdot 3 \cdot 31 \cdot 37$, the Frobenius trace $a_1(Q, p)$ is determined by*

$$a_1(Q, p) = \frac{a_1(X, p) - a_1(\mathrm{Jac}(X_A), p) - 2\, a_1(E_{\mathrm{res}}, p) - 3\, a_1(E, p)}{3},$$

where $\#X(\mathbb{F}_p) = 24\, N_{\mathrm{split}}(p) + 12(2 + c(p)) + 6$, with $N_{\mathrm{split}}(p)$ the number of $y \in \mathbb{F}_p \setminus \{0, 1\}$ for which $c(y) \neq 0$ and $\Pi(\lambda, y)$ splits completely, and $c(p) = \#\{y \in \mathbb{F}_p : c(y) = 0\}$. The first fifteen values are:

p	$a_1(\mathrm{Jac}(X_A))$	$a_1(E_{\mathrm{res}})$	$a_1(E)$	$a_1(Q)$
5	1	-2	-2	-5
7	3	-5	-1	-4
11	6	-3	-5	-5
13	0	-2	-2	-6
17	3	0	0	-5
19	2	-6	0	-4
23	3	0	2	-5
29	-1	-4	6	-3
41	10	-5	-9	1
43	-4	6	2	-12
47	6	-9	-9	-9
53	-8	-5	1	-7
59	-3	-6	8	7
61	4	8	-8	-8
67	1	-4	8	-9

All computed values satisfy the Weil bound $|a_1(Q,p)| \le 6\sqrt{p}$ *and the divisibility* $3 \mid (a_1(X,p) - a_1(\mathrm{Jac}(X_A),p) - 2\,a_1(E_{\mathrm{res}},p) - 3\,a_1(E,p))$*, providing a numerical consistency check for the isogeny decomposition of Theorem 4.7.*

Proof. For the regular S_4-closure $\pi\colon X \to \mathbb{P}^1$, the fiber above an unramified rational point $y \in \mathbb{P}^1(\mathbb{F}_p)$ consists of $|S_4| = 24$ geometric points on which Frobenius acts by left multiplication. Since left multiplication by a non-identity element of S_4 has no fixed points, the fiber contributes 24 rational points if and only if the Frobenius is trivial, i.e. $\Pi(\lambda, y)$ splits completely; otherwise the fiber contributes 0. The branch locus consists of $y = 0$, $y = 1$ (transposition inertia, $|S_4/\langle\sigma\rangle| = 12$ points each), the roots of $c(y)$ (transposition inertia, 12 points each), and $y = \infty$ (4-cycle inertia, $|S_4/\langle\tau\rangle| = 6$ points).

By Theorem 6.3,

$$3\,a_1(Q,p) = a_1(X,p) - a_1(\mathrm{Jac}(X_A),p) - 2\,a_1(E_{\mathrm{res}},p) - 3\,a_1(E,p).$$

The traces $a_1(E,p)$ and $a_1(E_{\mathrm{res}},p)$ are obtained by point counting on the explicit Weierstrass models of E and E_{res}, while $a_1(\mathrm{Jac}(X_A),p)$ is computed from the hyperelliptic model $w^2 = -y(y-1)c(y)$. Divisibility by 3 and the Weil bound are verified for all primes $p < 120$ with $p \nmid 2 \cdot 3 \cdot 31 \cdot 37$. □

6.5 Conductor and reduction

Proposition 6.10. *The discriminant of the hyperelliptic model* $f(y) = -y(y-1)c(y)$ *of* X_A *is*

$$\mathrm{Disc}(f) = -2^{20} \cdot 3^{15} \cdot 31^2 \cdot 37.$$

The set of primes of bad reduction for X_A *is contained in* $\{2, 3, 31, 37\}$.

Proof. The roots of $f(y) = -y(y-1)c(y)$ are $0, 1, \beta_1, \beta_2, \beta_3$, where β_i are the roots of $c(y)$. For leading coefficient $a_5 = -256$,

$$\mathrm{Disc}(f) = a_5^8 \prod_{i<j} (r_i - r_j)^2.$$

The product of pairwise differences factors as

$$\prod_{i<j} (r_i - r_j)^2 = (0-1)^2 \cdot \prod_i (0 - \beta_i)^2 \cdot \prod_i (1 - \beta_i)^2 \cdot \prod_{i<j} (\beta_i - \beta_j)^2.$$

By Vieta's formulas for $c(y) = 256\prod_i(y-\beta_i)$:

$$\prod_i \beta_i = -\frac{32}{256} = -\frac{1}{8}, \qquad \prod_i (1-\beta_i) = \frac{c(1)}{256} = \frac{864}{256} = \frac{27}{8}.$$

Using $\mathrm{Disc}(c) = 256^4 \prod_{i<j}(\beta_i-\beta_j)^2 = -3^9\cdot 31^2\cdot 37$ (Proposition 2.3),

$$\prod_{i<j}(r_i-r_j)^2 = 1\cdot\frac{1}{64}\cdot\frac{729}{64}\cdot\frac{-3^9\cdot 31^2\cdot 37}{256^4} = \frac{-3^{15}\cdot 31^2\cdot 37}{2^{44}}.$$

Therefore $\mathrm{Disc}(f) = (-256)^8\cdot(-3^{15}\cdot 31^2\cdot 37)/2^{44} = 2^{64}\cdot(-3^{15}\cdot 31^2\cdot 37)/2^{44} = -2^{20}\cdot 3^{15}\cdot 31^2\cdot 37$. □

Theorem 6.11. *The reduction types of the genus-2 curve X_A at the bad primes are as follows:*

(i) At $p = 31$: the polynomial $f(y) \bmod 31$ has exactly one double root (at $y \equiv 12$), giving semistable reduction with one ordinary double point. The conductor exponent is $f_{31} = 1$.

(ii) At $p = 37$: the polynomial $f(y) \bmod 37$ has exactly one double root (at $y \equiv 6$), giving semistable reduction with one ordinary double point. The conductor exponent is $f_{37} = 1$.

(iii) At $p = 3$: the branch cubic collapses to $c(y) \equiv (y-1)^3 \pmod 3$, so

$$f(y) \equiv 2y(y-1)^4 \pmod 3,$$

and the special fiber has a non-split tacnode at $(1,0)$ of delta invariant 2: the two local branches are conjugate over $\mathbb{F}_9$ and meet to second order. Moreover, the three non-rational branch points form a single depth-3/2 cluster above $y = 1$, so X_A does not satisfy the semistability criterion of [17, Definition 1.8 and Theorem 1.9]; in particular $X_A/\mathbb{Q}_3$ is not semistable.

(iv) The conductor of $\mathrm{Jac}(X_A)$ has the form

$$N_A = 2^{f_2}\cdot 3^{f_3}\cdot 31\cdot 37, \qquad f_2 \geq 1,$$

and Theorem 6.15 identifies the wild part at 3 as $\delta_3 = 1$.

Proof. For (i), one computes $c(y) \equiv 8(y-6)(y-12)^2 \pmod{31}$, so

$$f(y) \equiv -8y(y-1)(y-6)(y-12)^2 \pmod{31},$$

with simple roots $0, 1, 6$ and double root 12. The double root produces a single ordinary node on the special fiber. For a tame prime with a single node, Liu's analysis [18] gives $f_p = 1$.

For (ii), modulo 37 the branch cubic factors as $c(y) \equiv -3(y-6)^2(y-14) \pmod{37}$, giving

$$f(y) \equiv 3y(y-1)(y-6)^2(y-14) \pmod{37}.$$

and the same argument gives $f_{37} = 1$.

For (iii), the branch cubic satisfies $c(y) \equiv (y-1)^3 \pmod 3$, so the special fiber of $w^2 = f(y)$ over $\mathbb{F}_3$ is $w^2 = 2y(y-1)^4$. The singularity at $(1,0)$ is a non-split tacnode: substituting $y = 1+t$ gives $w^2 = 2t^4(1+t)$, whose two local branches $w = \pm\sqrt{2}\,t^2\sqrt{1+t}$ are conjugate over $\mathbb{F}_9$ (since $\sqrt{2} \notin \mathbb{F}_3$) and meet to second order, giving delta invariant 2. Proposition 6.13(i) shows that the three cubic branch points form a depth-3/2 cluster requiring a totally ramified cubic extension of $\mathbb{Q}_3$ to separate. Since the semistability criterion of [17, Definition 1.8] requires the root field $K(\mathcal{R})$ to have ramification degree at most 2, Theorem 1.9 of loc. cit. implies that $X_A/\mathbb{Q}_3$ is not semistable.

For (iv), modulo 2 the polynomial f reduces to $y^2(y+1)^2$, which is a perfect square, so $f_2 \geq 1$. The wild part at 3 is identified in Theorem 6.15. □

Proposition 6.12. *The nodal singularities of the special fibers of X_A at the two semistable primes have opposite splitting types:*

(i) At $p = 31$, the node is split: the tangent directions at the double root $y \equiv 12$ are rational over $\mathbb{F}_{31}$. The normalization of the nodal special fiber is the elliptic curve $v^2 = -8y(y-1)(y-6)$ over $\mathbb{F}_{31}$.

(ii) At $p = 37$, the node is non-split: the tangent directions at the double root $y \equiv 6$ are conjugate over $\mathbb{F}_{37^2}$. The normalization is the elliptic curve $v^2 = 3y(y-1)(y-14)$ over $\mathbb{F}_{37}$.

In both cases, the Jacobian $\mathrm{Jac}(X_A)$ acquires toric rank 1 at each prime, with the abelian part of the Néron model equal to the Jacobian of the normalization.

Proof. Write $f(y) = (y-a)^2 g(y)$ at the double root $y \equiv a \pmod p$. The tangent cone of $w^2 = f(y)$ at the node is $v^2 = g(a)$ (after $v = w/(y-a)$), which splits over $\mathbb{F}_p$ if and only if $g(a) \in (\mathbb{F}_p^\times)^2$.

At $p = 31$, $a = 12$: we have $g(y) = -8y(y-1)(y-6)$ and

$$g(12) \equiv -8 \cdot 12 \cdot 11 \cdot 6 \equiv 19 \equiv 9^2 \pmod{31},$$

so the node is split.

At $p = 37$, $a = 6$: here $g(y) = 3y(y-1)(y-14)$ and

$$g(6) \equiv 3 \cdot 6 \cdot 5 \cdot (-8) \equiv 20 \pmod{37}.$$

The Legendre symbol $\left(\frac{20}{37}\right) = \left(\frac{5}{37}\right) = -1$, so $g(6)$ is a quadratic non-residue and the node is non-split.

In both cases $v^2 = g(y)$ is an elliptic curve over $\mathbb{F}_p$, and the single node gives toric rank 1 in the Néron model [18]. □

Proposition 6.13. *The p-adic cluster pictures of the roots of $f(y) = -y(y-1)c(y)$ at the two wild primes are as follows.*

(i) At $p = 3$: the Newton polygon of the translated polynomial $c(1+3t)/27 = 256t^3 + 393t^2 + 195t + 32$ centered at the collapsed root $y = 1$ shows that the three roots β_i of $c(y)$ satisfy $v_3(\beta_i - 1) = 1$ and $v_3(\beta_i - \beta_j) = 3/2$ for all $i \neq j$. The cluster picture is

$$\{0\} \;\Big|\; \underbrace{\{1\} \;\Big|\; \underbrace{\{\beta_1, \beta_2, \beta_3\}}_{\text{depth } 3/2}}_{\text{depth } 1},$$

where the roots β_i are conjugate over $\mathbb{Q}_3$ and require a totally ramified cubic extension to separate. Since this extension has degree $3 = p$, it contributes a nontrivial Swan conductor $\delta_3 > 0$.

(ii) At $p = 2$: the Newton polygon of $c(y) = 256y^3 + 411y^2 + 165y + 32$ has vertices at $(0, v_2(32)) = (0, 5)$, $(1, v_2(165)) = (1, 0)$, $(2, v_2(411)) = (2, 0)$, $(3, v_2(256)) = (3, 8)$, giving slopes $-5, 0, 8$. (With the convention that the Newton polygon slope between (i, v_i) and $(i+1, v_{i+1})$ is $v_{i+1} - v_i$, each root β satisfies $v_2(\beta) = -$slope of the corresponding segment.) Each segment has length 1, so by Hensel's lemma all three roots lie in $\mathbb{Q}_2$. Denote these roots $\beta_1, \beta_2, \beta_3$. Since $v_2(c(1)) = 5$ and $v_2(c'(1)) = 0$, one has $v_2(\beta_2 - 1) = 5$. The cluster picture is

$$\underbrace{\{0, \beta_1\}}_{\text{depth } 5} \;\Big|\; \underbrace{\{1, \beta_2\}}_{\text{depth } 5} \;\Big|\; \{\beta_3\},$$

with two balanced clusters of depth 5 and one isolated root of valuation -8.

Proof. For (i), the substitution $y = 1 + 3t$ gives $c(1 + 3t) = 27(256t^3 + 393t^2 + 195t + 32) = 27h(t)$. A further substitution $t = 1 + u$ yields

$$h(1 + u) = 256u^3 + 1161u^2 + 1749u + 876,$$

with 3-adic valuations $v_3(876) = 1$, $v_3(1749) = 1$, $v_3(1161) = 3$, $v_3(256) = 0$. The Newton polygon has a single segment of slope $-1/3$ from $(0, 1)$ to $(3, 0)$, so all three roots satisfy $v_3(t_i - 1) = 1/3$. In particular $v_3(\beta_i - 1) = 1 + v_3(t_i) = 1$, and

$$v_3(\beta_i - \beta_j) = 1 + v_3(t_i - t_j).$$

The pairwise distances follow from $\mathrm{Disc}(h) = -3^3 \cdot 31^2 \cdot 37$, giving

$$\textstyle\sum_{i<j} v_3(t_i - t_j) = \frac{1}{2}\big(v_3(\mathrm{Disc}(h)) - 4 \cdot v_3(256)\big) = \frac{3}{2},$$

and by symmetry each $v_3(t_i - t_j) = 1/2$, whence $v_3(\beta_i - \beta_j) = 3/2$.

For (ii), the 2-adic Newton polygon of $c(y)$ has vertices $(0, 5)$, $(1, 0)$, $(2, 0)$, $(3, 8)$, giving slopes -5, 0, 8. The root β_1 with $v_2(\beta_1) = 5$ clusters with 0; the root β_2 with $v_2(\beta_2) = 0$ satisfies $v_2(\beta_2 - 1) = 5$ by Newton's method applied to $c(1 + h) = 0$, since $v_2(c(1)) = 5$ and $v_2(c'(1)) = 0$; the root β_3 with $v_2(\beta_3) = -8$ is isolated. □

Proposition 6.14. *Let $F_w = \mathbb{Q}_3(\beta_1)$, where β_1 is any root of the branch cubic $c(y)$. Then:*

(i) $F_w/\mathbb{Q}_3$ is totally ramified of degree 3;

(ii) the local discriminant exponent is $v_3(\Delta_{F_w/\mathbb{Q}_3}) = 3$;

(iii) the finite branch roots of $f(y) = -y(y - 1)c(y)$ form exactly three $G_{\mathbb{Q}_3}$-orbits, namely

$$\{0\}, \qquad \{1\}, \qquad \{\beta_1, \beta_2, \beta_3\}.$$

Proof. By Proposition 6.13(i), the roots β_i are conjugate over $\mathbb{Q}_3$ and satisfy $v_3(\beta_i - 1) = 1$ and $v_3(\beta_i - \beta_j) = 3/2$. In particular the denominator 2 in the pairwise distances forces a nontrivial ramified extension, and the same proposition shows that a totally ramified cubic extension is required to separate the three roots. Thus $[F_w : \mathbb{Q}_3] = 3$ and $F_w/\mathbb{Q}_3$ is totally ramified, proving (i). The orbit statement (iii) is the same cluster-theoretic description, together with the rationality of the roots 0 and 1.

For (ii), Proposition 2.3 gives $\mathrm{Disc}(F/\mathbb{Q}) = -3^3 \cdot 37$. Since $F_w/\mathbb{Q}_3$ is totally ramified of degree 3, the prime 3 has a unique extension to F and residue degree 1. Therefore the exponent of 3 in the global field discriminant equals the local discriminant exponent:

$$v_3(\mathrm{Disc}(F/\mathbb{Q})) = f_w\, v_3(\Delta_{F_w/\mathbb{Q}_3}) = 1 \cdot v_3(\Delta_{F_w/\mathbb{Q}_3}),$$

so $v_3(\Delta_{F_w/\mathbb{Q}_3}) = 3$. □

Theorem 6.15. *The wild part of the conductor of* $\mathrm{Jac}(X_A)$ *at $p = 3$ is exactly*

$$\delta_3 = 1.$$

Proof. Let

$$R = \{0, 1, \beta_1, \beta_2, \beta_3\}$$

be the set of finite branch roots of the odd-degree model

$$w^2 = -256 \prod_{r \in R} (y - r).$$

The branch polynomial has odd degree 5, the residue characteristic is $p = 3 \neq 2$, and the roots are distinct in $\overline{\mathbb{Q}}_3$ because $\mathrm{Disc}(f) \neq 0$. Therefore the conductor formula of Dokchitser, Dokchitser, Maistret, and Morgan (Math. Ann. 385 (2023), Theorem 11.3) applies exactly in this setting: for an odd-degree hyperelliptic curve, the wild conductor is expressed as a sum over the $G_{\mathbb{Q}_3}$-orbits on the finite branch set R, with no extra contribution from the point at infinity. Since the scalar factor -1 is a 3-adic unit, it does not affect the extension fields $\mathbb{Q}_3(r)$ or the conductor contribution of any root. Thus

$$\delta_3 = \sum_{r \in R//G_{\mathbb{Q}_3}} \Big(v_3(\Delta_{\mathbb{Q}_3(r)/\mathbb{Q}_3}) - [\mathbb{Q}_3(r) : \mathbb{Q}_3] + f(\mathbb{Q}_3(r)/\mathbb{Q}_3) \Big),$$

where $R//G_{\mathbb{Q}_3}$ denotes a set of $G_{\mathbb{Q}_3}$-orbit representatives and $f(\mathbb{Q}_3(r)/\mathbb{Q}_3)$ is the residue degree.

By Proposition 6.14, the $G_{\mathbb{Q}_3}$-orbits on R are $\{0\}$, $\{1\}$, and $\{\beta_1, \beta_2, \beta_3\}$. The rational roots 0 and 1 contribute

$$v_3(\Delta_{\mathbb{Q}_3/\mathbb{Q}_3}) - [\mathbb{Q}_3 : \mathbb{Q}_3] + f(\mathbb{Q}_3/\mathbb{Q}_3) = 0 - 1 + 1 = 0.$$

Equivalently, the cluster picture contributes two rational singleton orbits and one cubic orbit centered 3-adically at 1, with pairwise distances $v_3(\beta_i - \beta_j) = 3/2$. All of the wild contribution is therefore concentrated in that single cubic orbit. For the cubic orbit, Proposition 6.14 gives $[F_w : \mathbb{Q}_3] = 3$, $f(F_w/\mathbb{Q}_3) = 1$, and $v_3(\Delta_{F_w/\mathbb{Q}_3}) = 3$. Therefore the cubic orbit contributes

$$v_3(\Delta_{F_w/\mathbb{Q}_3}) - [F_w : \mathbb{Q}_3] + f(F_w/\mathbb{Q}_3) = 3 - 3 + 1 = 1,$$

so the full sum is

$$\delta_3 = (0 - 1 + 1) + (0 - 1 + 1) + (3 - 3 + 1) = 1.$$

□

Remark 6.16. The conductor exponent f_2 remains undetermined. At $p = 2$, the special fiber of the naive hyperelliptic model is $w^2 \equiv y^2(y-1)^2 \pmod 2$, a non-reduced curve, so the minimal regular model over $\mathbb{Z}_2$ requires a nontrivial sequence of blowups whose combinatorics depend on the full 2-adic cluster picture. Since all three roots of $c(y)$ lie in $\mathbb{Q}_2$ (Proposition 6.13(ii)), the stable reduction is in principle computable, but the two balanced clusters at depth 5 make the computation substantially more involved than the wild-part calculation at $p = 3$.

6.6 Torsion and Galois representations

Theorem 6.17. *The* 2*-torsion field of* $\mathrm{Jac}(X_A)$ *coincides with the splitting field of the branch cubic:*

$$\mathbb{Q}(\mathrm{Jac}(X_A)[2]) = L.$$

*The mod-*2 *Galois representation*

$$\bar{\rho}_2 \colon G_{\mathbb{Q}} \longrightarrow \mathrm{Sp}(4, \mathbb{F}_2) \cong S_6$$

has image isomorphic to S_3*, of order* 6 *inside* $|\mathrm{Sp}(4, \mathbb{F}_2)| = 720$.

Proof. For a genus-2 curve $C : w^2 = f(y)$ with $\deg f = 5$, the nonzero elements of $\mathrm{Jac}(C)[2]$ are represented by divisor classes of the form

$$[(r_i, 0) - (r_j, 0)],$$

where r_i, r_j are Weierstrass points; equivalently, the 2-torsion field is the splitting field of f [19, Ch. 2]. In our case

$$f(y) = -y(y-1)c(y)$$

has Weierstrass points above 0, 1, and the three roots $\beta_1, \beta_2, \beta_3$ of $c(y)$. Since 0 and 1 are rational, adjoining all 2-torsion points amounts to adjoining the cubic branch points β_i. Hence

$$\mathbb{Q}(\mathrm{Jac}(X_A)[2]) = \mathbb{Q}(\beta_1, \beta_2, \beta_3) = L.$$

Let $W_0 = \langle D_0, D_1 \rangle$, where $D_0 = [(0,0) - \infty]$ and $D_1 = [(1,0) - \infty]$. Then W_0 is fixed pointwise by $G_{\mathbb{Q}}$. In the quotient $\mathrm{Jac}(X_A)[2]/W_0$, the classes $\bar{D}_{\beta_1}, \bar{D}_{\beta_2}$ form an $\mathbb{F}_2$-basis and satisfy $\bar{D}_{\beta_3} = \bar{D}_{\beta_1} + \bar{D}_{\beta_2}$. Therefore a 3-cycle $\sigma = (123)$ and a transposition $\tau = (12)$ act by

$$\sigma \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \tau \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

on $\mathrm{Jac}(X_A)[2]/W_0$. These matrices generate $\mathrm{GL}(2, \mathbb{F}_2) \cong S_3$, so the mod-2 image is exactly S_3. Since the Weil pairing is Galois invariant, this image embeds into $\mathrm{Sp}(4, \mathbb{F}_2)$ as a subgroup of order 6. □

Corollary 6.18. *Theorem 5.4(iv) identifies L as the unique cubic subextension of $K_{\mathfrak{p}_3^2}/K$. The 2-torsion of $\mathrm{Jac}(X_A)$ is therefore controlled by the ray class group $\mathrm{Cl}_{\mathfrak{p}_3^2}(K) \cong C_{24}$, and its absolute discriminant is $\mathrm{Disc}(L/\mathbb{Q}) = 3^7 \cdot 37^3$ (Theorem 5.2(iv)). The same extension $L/\mathbb{Q}$ also determines the weight-1 Artin representation of Corollary 5.6.*

Proof. Combine Theorem 6.17 with Theorem 5.4(iv). □

6.7 The $\mathbb{F}_2[S_3]$-module structure of the 2-torsion

Theorem 6.17 identifies the 2-torsion field as the splitting field L of $c(y)$ and shows that the mod-2 image is S_3. We now refine this to a complete description of the module structure and the resulting embedding.

Theorem 6.19. *As an $\mathbb{F}_2[\mathrm{Gal}(L/\mathbb{Q})]$-module, the 2-torsion admits a decomposition*

$$\mathrm{Jac}(X_A)[2] \cong W_0 \oplus W_1,$$

where $W_0 = \langle D_0, D_1 \rangle \cong \mathbb{F}_2^2$ is the trivial module spanned by the rational Weierstrass divisor classes and $W_1 \cong \mathrm{St}_2$ is the standard two-dimensional representation of $S_3 \cong \mathrm{GL}(2, \mathbb{F}_2)$. Both W_0 and W_1 are nondegenerate (hyperbolic) subspaces for the Weil pairing, and the decomposition is a symplectic orthogonal sum:

$$\big(\mathrm{Jac}(X_A)[2], e_2\big) \cong (W_0, e_2|_{W_0}) \perp (W_1, e_2|_{W_1}).$$

Consequently, the mod-2 Galois image lies in the subgroup

$$\bar{\rho}_2(G_{\mathbb{Q}}) \cong S_3 \hookrightarrow \{1\} \times \mathrm{Sp}(W_1) \subset \mathrm{Sp}(W_0) \times \mathrm{Sp}(W_1) \subset \mathrm{Sp}(4, \mathbb{F}_2),$$

acting trivially on the rational hyperbolic plane W_0 and faithfully on the irrational hyperbolic plane W_1.

Proof. For the genus-2 curve $w^2 = f(y)$ with $\deg f = 5$, the 2-torsion $\mathrm{Jac}(X_A)[2] \cong (\mathbb{Z}/2\mathbb{Z})^4$ is generated by the divisor classes $D_i = [(r_i, 0) - \infty]$ for the five roots r_i of f, subject to $\sum D_i = 0$ [19]. The roots are $0, 1, \beta_1, \beta_2, \beta_3$. Taking $\{D_0, D_1, D_{\beta_1}, D_{\beta_2}\}$ as a basis, the Galois group $S_3 = \mathrm{Gal}(L/\mathbb{Q})$ fixes D_0 and D_1, and permutes $\{D_{\beta_1}, D_{\beta_2}, D_{\beta_3}\}$ with $D_{\beta_3} = D_0 + D_1 + D_{\beta_1} + D_{\beta_2}$.

For a genus-2 Jacobian in a Weierstrass basis, the Weil pairing on 2-torsion satisfies $e_2(D_i, D_j) = -1$ for $i \neq j$ [19]. Hence the Gram matrix in additive $\mathbb{F}_2$-notation with respect to $\{D_0, D_1, D_{\beta_1}, D_{\beta_2}\}$ is

$$M = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

Consider the symplectic basis $e_1 = D_0,\ f_1 = D_1,\ e_2 = D_{\beta_1} + D_{\beta_2},\ f_2 = D_0 + D_1 + D_{\beta_2}$. A direct computation gives

$$\langle e_1, f_1 \rangle = 1, \quad \langle e_2, f_2 \rangle = 1, \quad \langle e_i, e_j \rangle = \langle f_i, f_j \rangle = \langle e_i, f_j \rangle = 0 \quad (i \neq j),$$

exhibiting the standard symplectic form $J_2 \perp J_2$. The plane $W_0 = \langle e_1, f_1 \rangle$ is fixed by S_3, and its symplectic complement $W_1 = \langle e_2, f_2 \rangle$ carries the standard representation (the quotient action on $\{D_{\beta_1}, D_{\beta_2}, D_{\beta_3}\}$ modulo W_0 is the standard 2-dimensional representation of $S_3 \cong \mathrm{GL}(2, \mathbb{F}_2)$). Both summands are hyperbolic, and the decomposition is S_3-equivariant.

The subgroup of $\mathrm{Sp}(4, \mathbb{F}_2)$ preserving the orthogonal splitting $W_0 \perp W_1$ is $\mathrm{Sp}(W_0) \times \mathrm{Sp}(W_1) \cong S_3 \times S_3$. Since Theorem 6.17 identifies the quotient action on W_1 with the standard matrices above and S_3 acts trivially on W_0, the image is $\{1\} \times S_3$. □

Remark 6.20. The S_3-invariant hyperbolic splitting $W_0 \perp W_1$ is the unique such decomposition: the only 2-dimensional S_3-invariant nondegenerate subspace is $W_0 = \mathrm{Jac}(X_A)[2]^{G_\mathbb{Q}}$, and its symplectic complement W_1 is the unique S_3-stable complement. This rigidity contrasts with the situation for curves with fewer rational Weierstrass points, where the fixed space may be trivial and the Galois representation cannot be reduced in this manner.

6.8 Component groups at the semistable primes

Proposition 6.21. *The component groups and Tamagawa numbers of* $\mathrm{Jac}(X_A)$ *are as follows:*

(i) At $p = 31$ *(split node): the component group is* $\Phi_{31} \cong \mathbb{Z}/2\mathbb{Z}$*, and* $c_{31}(\mathrm{Jac}(X_A)) = 2$*.*

(ii) At $p = 37$ *(non-split node): the component group is trivial, and* $c_{37}(\mathrm{Jac}(X_A)) = 1$*.*

Proof. At each semistable prime, the stable model of X_A has a single ordinary double point (Theorem 6.11). Blowing up that node once produces the minimal regular model: the special fiber consists of the strict transform C_0 of the normalization, which is an elliptic curve, together with one exceptional divisor $E_0 \cong \mathbb{P}^1$. The exceptional divisor records the two tangent directions at the node.

For a one-nodal irreducible semistable curve, Raynaud's description of the Néron model identifies the identity component of the Jacobian with an extension of $\mathrm{Jac}(C_0)$ by a one-dimensional torus determined by those tangent directions; see [20, Chapter 10] and [18, Chapter 10]. If the node is split, the two tangent directions are rational and the torus is $\mathbb{G}_m$; the monodromy pairing has matrix [2], so the component group is

$$\Phi_p \cong \mathrm{coker}\big([2] \colon \mathbb{Z} \to \mathbb{Z}\big) \cong \mathbb{Z}/2\mathbb{Z}.$$

If the node is non-split, the tangent directions are exchanged by Frobenius and the toric part is the anisotropic norm-one torus $R^1_{\mathbb{F}_{p^2}/\mathbb{F}_p}\mathbb{G}_m$, whose Néron model is connected. In that case the component group is trivial.

By Proposition 6.12, the node at 31 is split, giving $\Phi_{31} \cong \mathbb{Z}/2\mathbb{Z}$ and $c_{31} = |\Phi_{31}(\mathbb{F}_{31})| = 2$. The node at 37 is non-split, giving $\Phi_{37} = 0$ and $c_{37} = 1$. □

6.9 Numerical verification

Proposition 6.22. *The following table records the point counts $\#X_A(\mathbb{F}_p)$ and $\#E_{\mathrm{res}}(\mathbb{F}_p)$ at the first twelve primes of good reduction for X_A, together with the Frobenius traces and the number of roots of $c(y)$ modulo p:*

p	$\#X_A(\mathbb{F}_p)$	$a_1(X_A)$	$\#E_{\mathrm{res}}(\mathbb{F}_p)$	$a_1(E_{\mathrm{res}})$	$\#\{y \in \mathbb{F}_p : c(y) = 0\}$
5	5	1	8	-2	0
7	5	3	13	-5	0
11	6	6	15	-3	1
13	14	0	16	-2	1
17	15	3	18	0	0
19	18	2	26	-6	1
23	21	3	24	0	0
29	31	-1	34	-4	0
41	32	10	47	-5	1
43	48	-4	38	6	1
47	42	6	57	-9	1
53	62	-8	59	-5	1

The characteristic polynomial of Frobenius $\chi_{13}(T) = T^4 - 2T^2 + 169$ is irreducible over $\mathbb{Q}$, in agreement with Proposition 6.1.

Proof. The point counts are computed by evaluating the Legendre symbol $(f(y)/p)$ for each $y \in \mathbb{F}_p$, together with the point at infinity. The characteristic polynomial is determined by $a_1 = p + 1 - \#X_A(\mathbb{F}_p)$ and $a_2 = (a_1^2 - p^2 - 1 + \#X_A(\mathbb{F}_{p^2}))/2$. Irreducibility of $\chi_{13}(T) = T^4 - 2T^2 + 169$ is shown in the proof of Proposition 6.1. □

Remark 6.23. The Hecke eigenvalues $a_p(f_c) = \#\{y \in \mathbb{F}_p : c(y) = 0\} - 1$ of the weight-1 form, computed from the last column of the table, match the values predicted by Theorem 5.7 at all primes listed. The values cycle through $\{-1, 0, 2\}$, corresponding to the Frobenius cycle types (123), (12), and 1 in S_3.

Remark 6.24. For a structured inventory of the companion scripts and generated files used in the numerical checks of Sections 5 and 6, see the reproducibility note recorded after Section 7.

7 Further directions

(1) Determine $\mathrm{End}^0(\mathrm{Jac}(X_A)_{\overline{\mathbb{Q}}})$ and the odd ℓ-adic monodromy of $\mathrm{Jac}(X_A)$.

(2) Construct an explicit model for $Y = X/C_4$ and determine the resulting geometry and L-function of the Prym threefold $Q = \mathrm{Prym}(Y/E_{\mathrm{res}})$.

(3) Proposition 6.13 determines the 2-adic and 3-adic cluster pictures, and Theorem 6.15 identifies the exact wild contribution $\delta_3 = 1$ at 3. Determining the full conductor at 3 and the exponent f_2 still requires a full stable-model computation or an independent automorphic input.

Reproducibility

The computational claims used in Sections 5 and 6 are accompanied by scripts in the source tree of the manuscript. The main files are:

- `point_count.py` in `artifacts/`, which reproduces the point counts of X_A over $\mathbb{F}_p$ and $\mathbb{F}_{p^2}$;
- `conductor_exact.py` in `artifacts/`, which checks the local conductor calculation at $p = 3$;
- `exp_branch_cubic_rayclass_modform_audit.py` in `scripts/`, which reproduces the class-group, ray-class, and Hecke-trace checks of Section 5;
- `exp_genus2_jacobian_audit.py` in `scripts/`, which audits the Frobenius polynomials of $\mathrm{Jac}(X_A)$;
- `generate_Q_traces_table.py` in `scripts/`, which generates the table for Q.

The companion note `REPRODUCIBILITY.md` records the role of each script and the generated outputs used in the manuscript. Database information from the LMFDB is used only as a consistency check and not as an input to any proof.

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Computations were performed with SageMath and Python. The LMFDB [11] was used as a consistency check for elliptic-curve metadata.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used OpenAI ChatGPT in order to improve the language, readability, and presentation of the manuscript, without generating the underlying mathematical ideas, arguments, or results. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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